

DATA 608 Meetup 4: Visualizing Associations and Time Series

George Hagstrom

Week 4 Summary

- Data Story 2 due this Sunday
- Discussion post on
- This week is all about associations
 - Two variables
 - Time Series
 - Several to Many Variables

Events Happening this Week!

Bachelor of Science in Information Systems

CAREER READINESS BOOTCAMP

CUNY | School of
Professional Studies

Kickstart your career with our BSIS Career Readiness Bootcamp!

This event is designed to equip you with the essential tools to stand out in today's competitive job market. Join us to learn about:

- **Resume Writing**
- **Interview Tips**
- **Networking Essentials**
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Take advantage of this opportunity to connect with the CIE staff, including your Career Advisor and Industry Specialist, who will guide you through the process of preparing for your future career.

To attend virtually, [register here](#). For in-person registration, [click here](#)!

Don't miss out—your next step toward success starts here!

Events Happening this Week

1. Career Readiness Bootcamp

- Wednesday Februar 19th, Room 407 319 West 31st Street

[In person link](#)

[Virtual link](#)

Events Happening This Week!

NEW YORK OPEN STATISTICAL PROGRAMMING MEETUP



SUBWAY STORIES: BUILDING AN AWARD-WINNING VISUALIZATION WITH TRANSIT DATA

Jediah Katz & Marc Zitelli

Software Engineer at Figma & Fmr. Political Consultant at Hamilton Campaign Network



In-Person & Virtual Event



Wednesday, February 19th, 2025

Doors open at 6:30 PM (America/New_York)

Talk is from 7:00 - 8:30 PM (America/New_York)



Visit nyhackr.org for more information.



Events Happening this Week

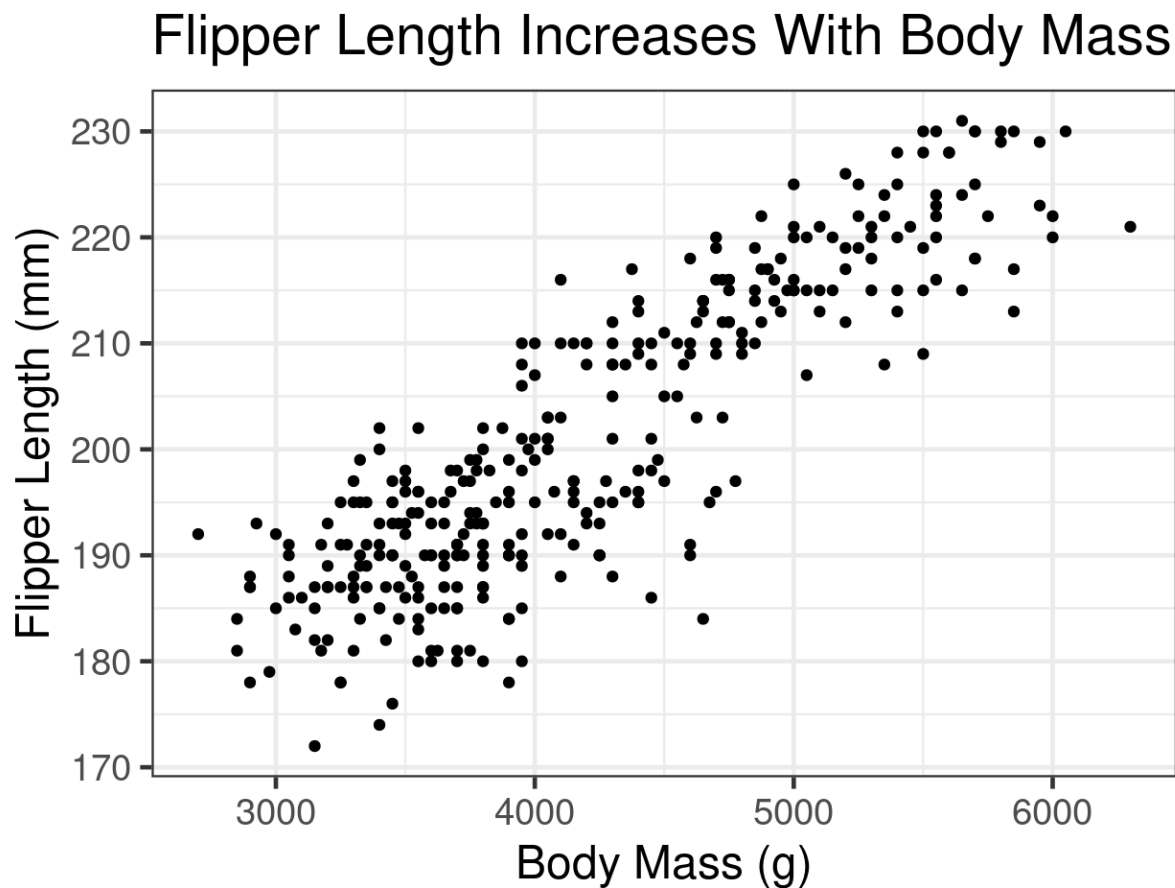
2. New York Open Statistical Computing Meetup

- 6:30-8:30+, Pless Hall NYU
- \$7 tickets must RSVP
- We have a group that attends these monthly meetups
- I won't be there this week not sure how the attendance will be

[Register by Clicking Here](#)

Scatterplots

- Shows how two variables are associated



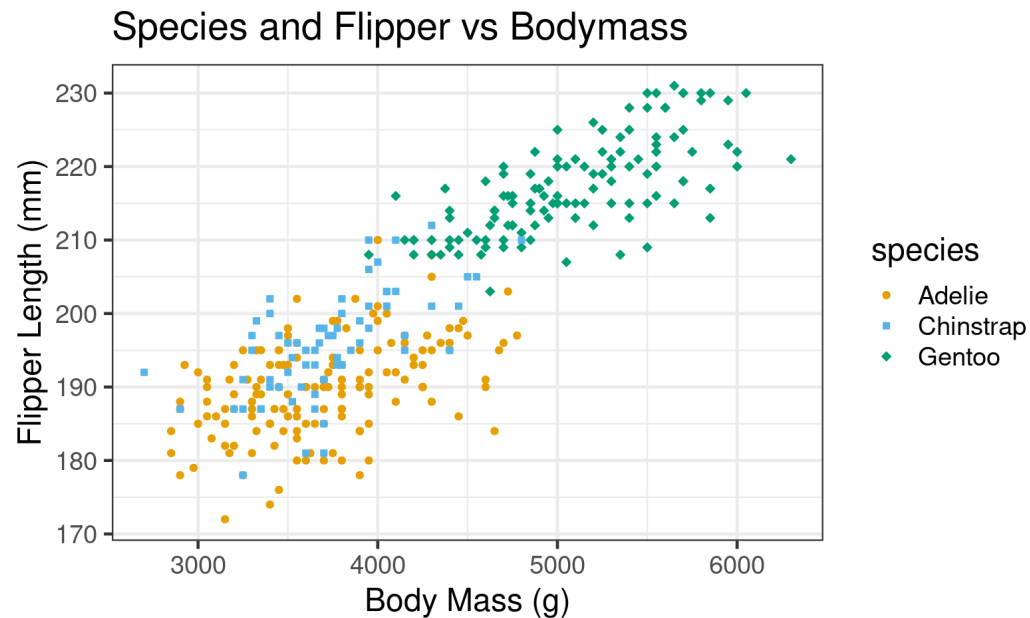
Scatterplots

- Shows how two variables are associated

```
1 library(tidyverse)
2 library(palmerpenguins)
3 library(ggplot2)
4
5 penguins |> ggplot(aes(x=body_mass_g, y=flipper_length_mm)) +
6   geom_point() +
7   theme_bw(base_size = 18) +
8   theme(aspect.ratio=0.67) +
9   labs( title = "Flipper Length Increases With Body Mass") +
10  xlab("Body Mass (g)") +
11  ylab("Flipper Length (mm)")
```

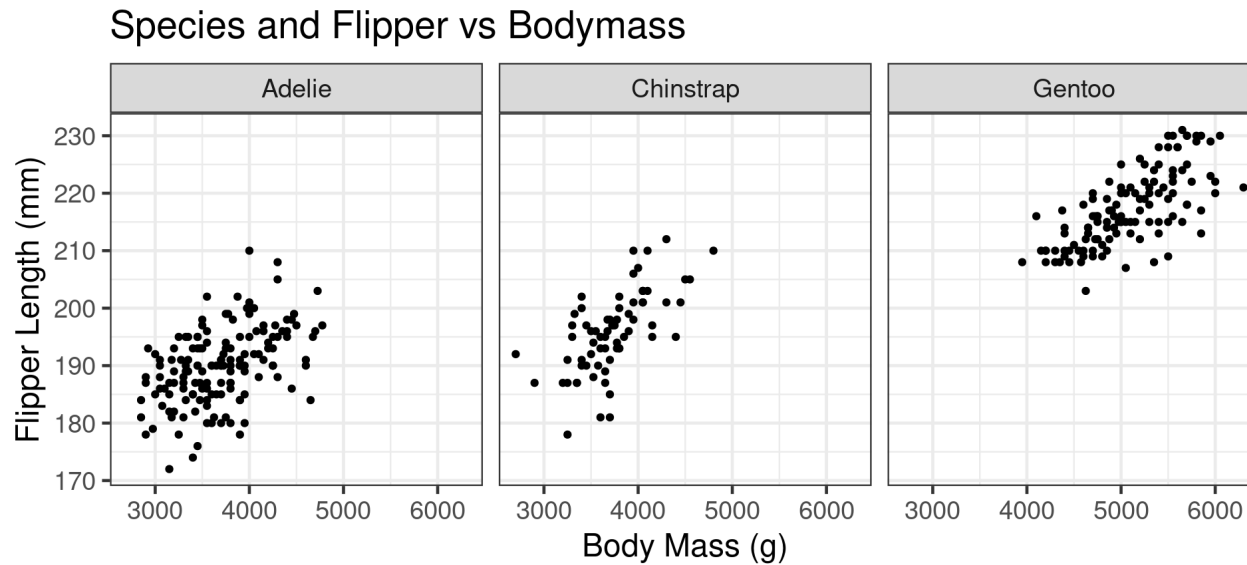

Multiple Types of Penguins

- Can add categorical information several ways:
 - Color/Shape (Same Graph)
 - Facets (Different Graphs)

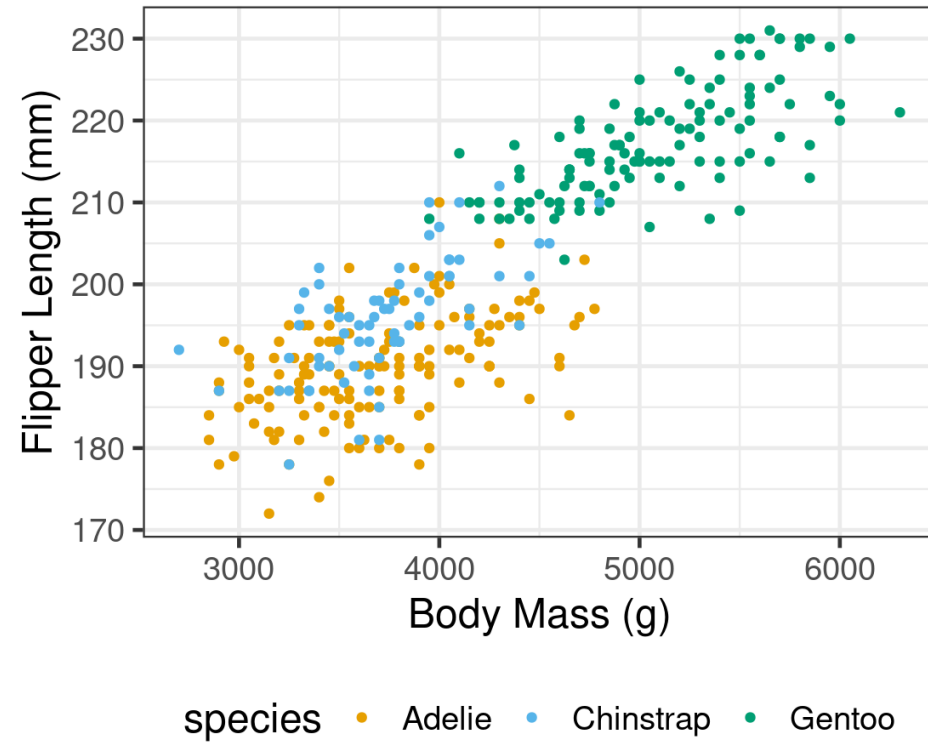
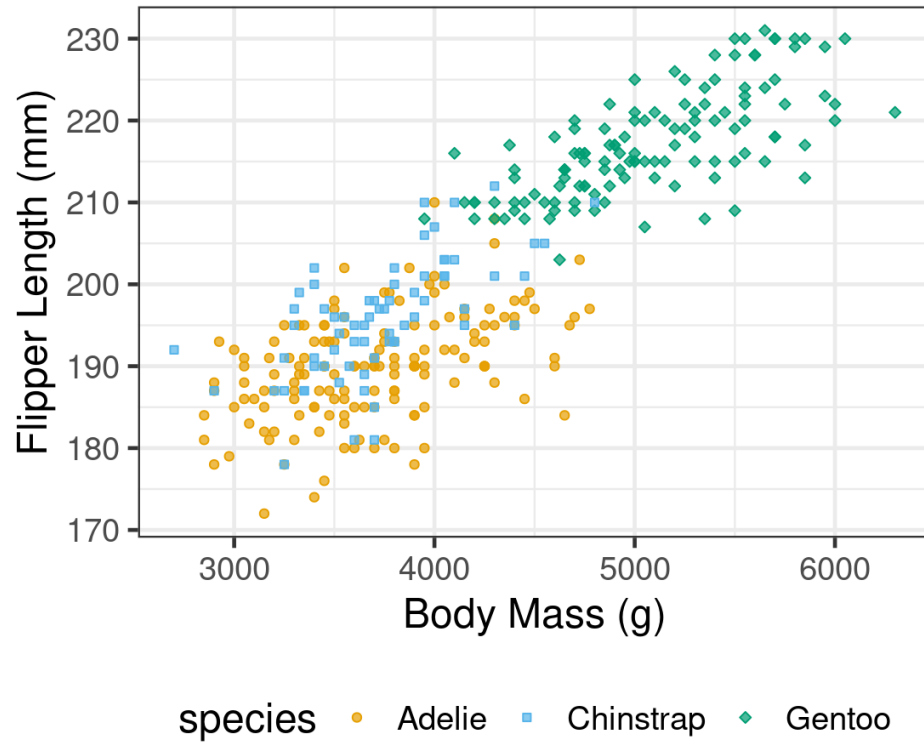


Multiple Types of Penguins

- Can add categorical information several ways:
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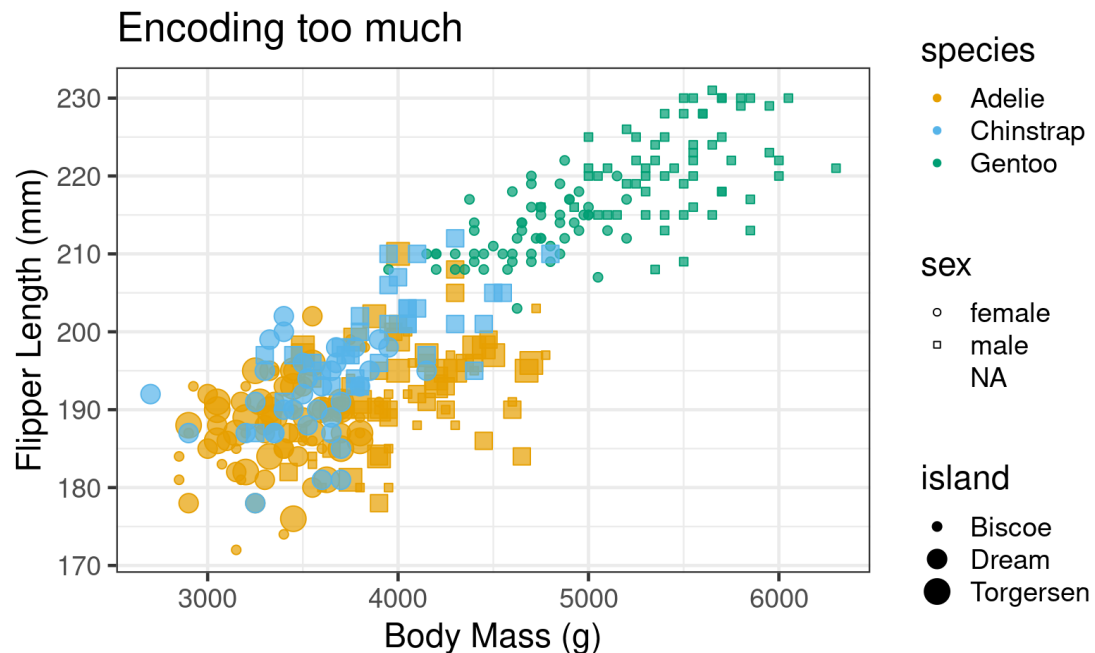
Redundant Coding



- Some people can differentiate better when the additional variable is coded multiple times

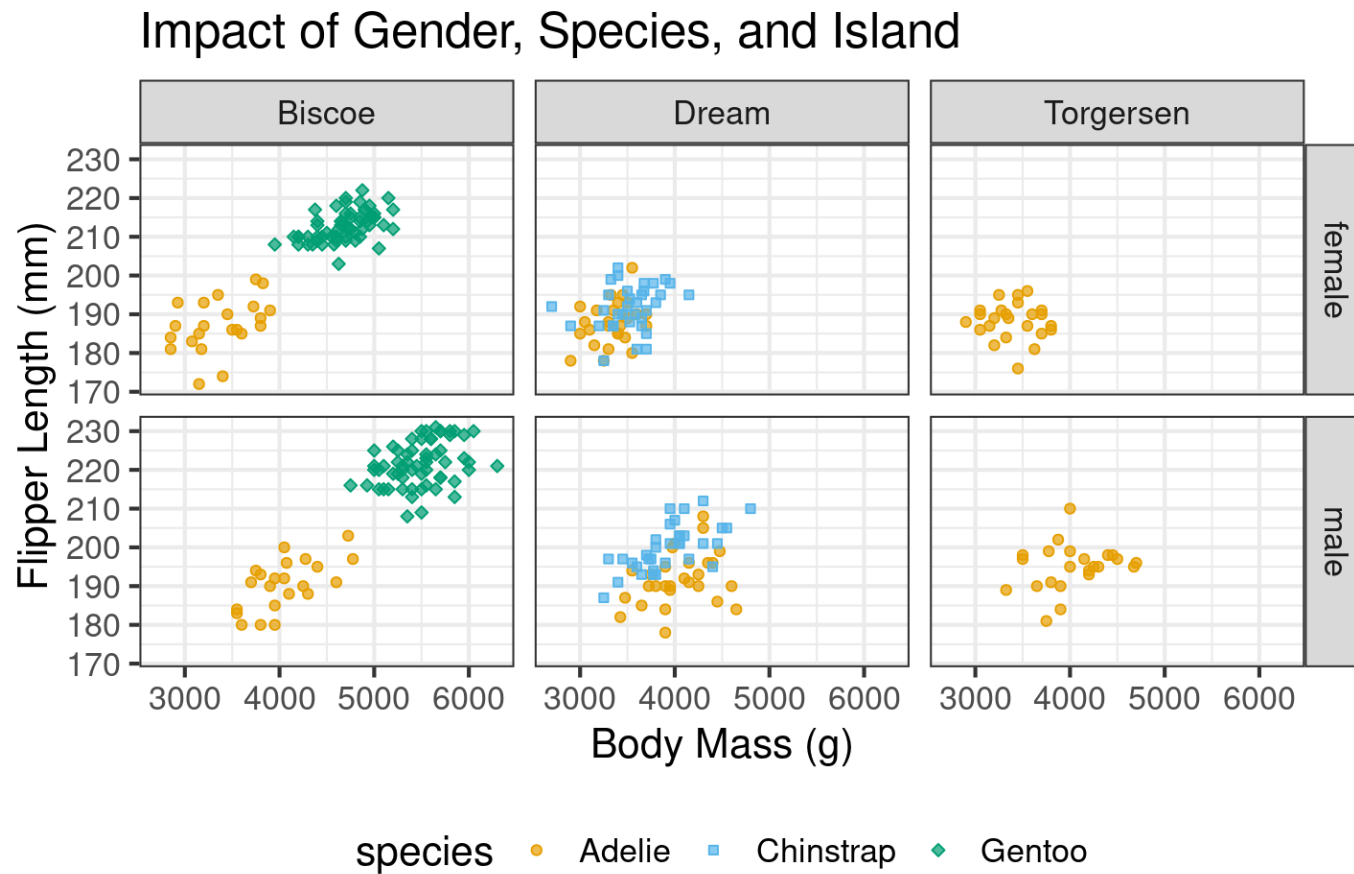
Limits on Channels

- Using multiple “channels” for different variables overloads the brain
- Practical limit is 1 categorical variable redundantly coded



Facets

- Facet Grid and Facet wrap allow more annotations

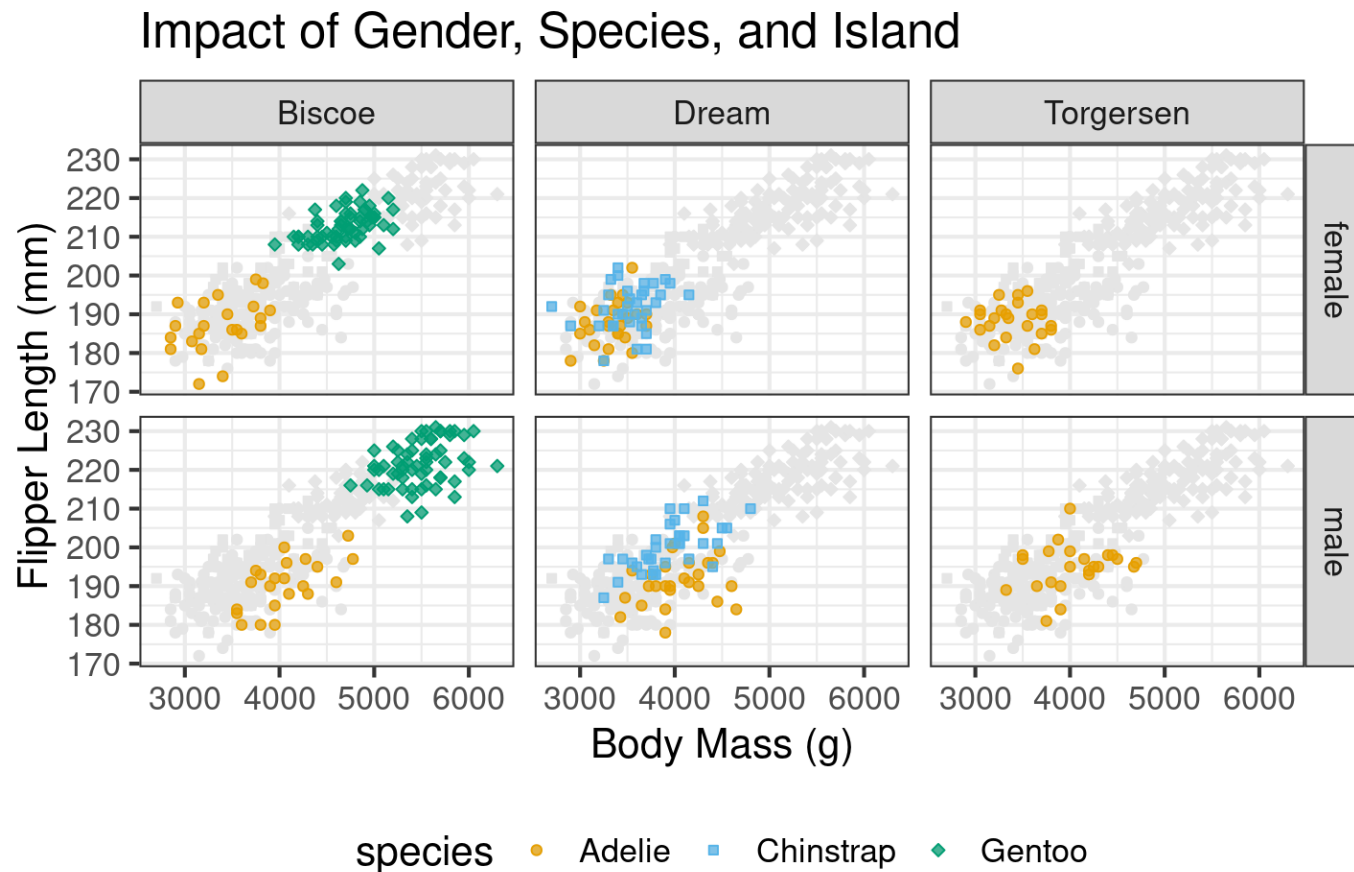


When to use Facets?

- Showing on the same plot is better because of direct comparisons unless:
 - You want to show several categories
 - Too many elements in category (6?)

Facilitating Comparisons with Facets

- Show all points on each facet



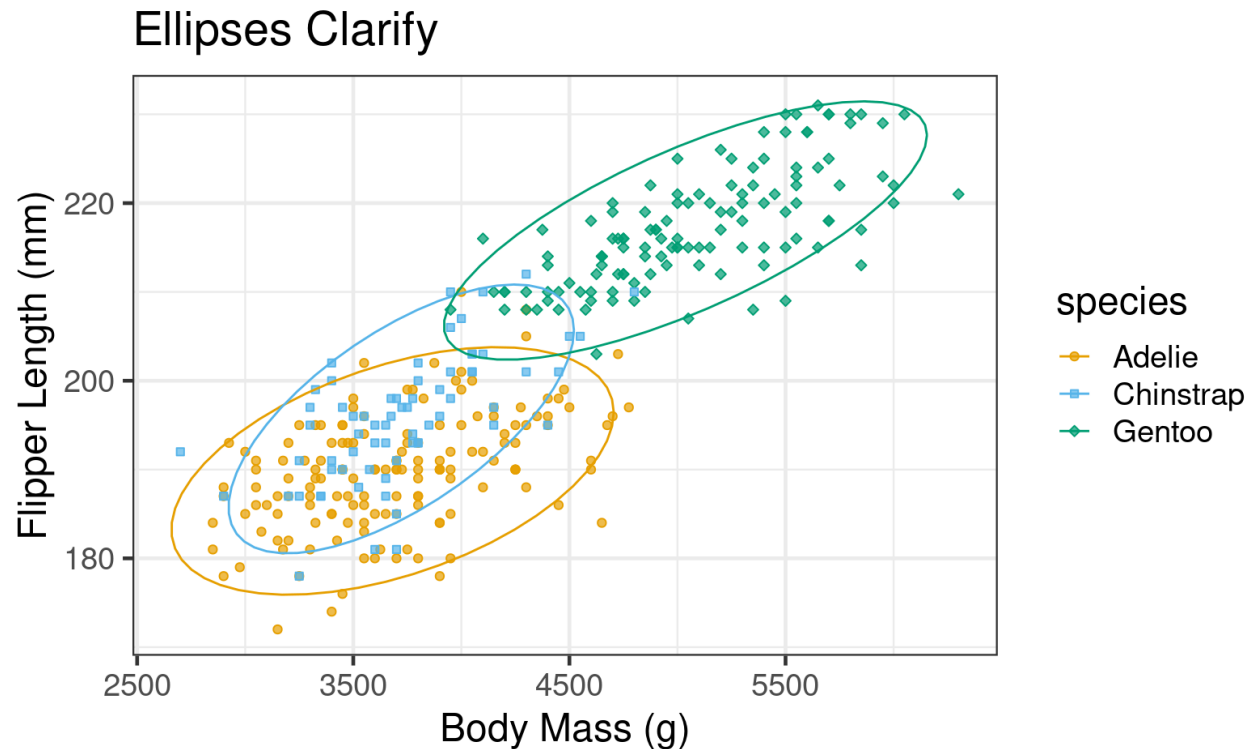
Facilitating Comparisons with Facets

```
1 penguins |> filter(sex %in% c("male","female")) |>
2   ggplot(aes(x=body_mass_g,
3             y=flipper_length_mm,
4             color=species,shape = species,
5             fill = species)) +
6   geom_point(data = transform(penguins,
7                               sex = NULL,
8                               island = NULL),
9             color = "grey90",
10            fill="grey90") +
11   facet_grid(sex ~ island) + ... other
```

- Add to old plot another `geom_point`
- `NULL` the facet categories
- Force a light color

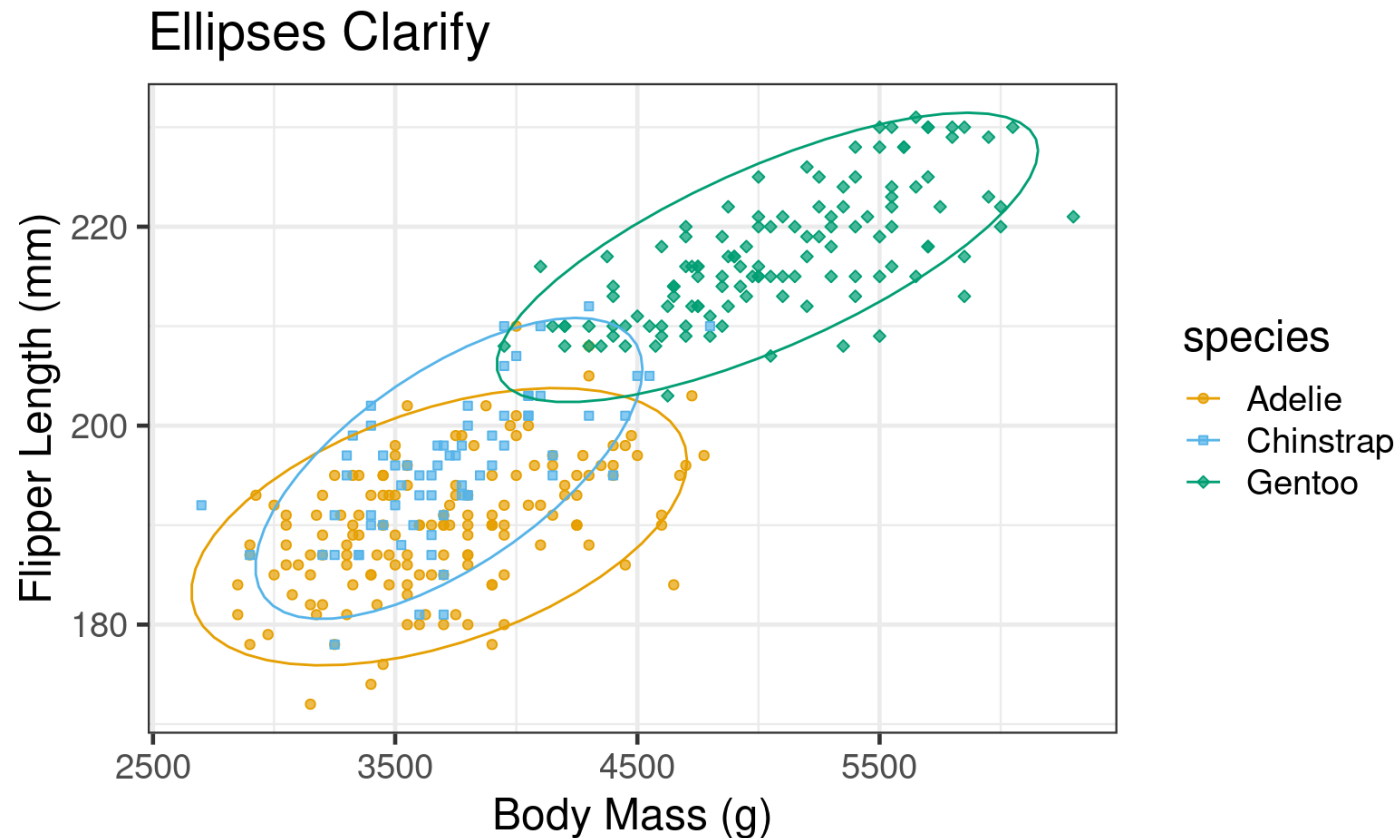
Scatterplot Annotations

- Reduce cognitive load by labeling directly
- `stat_ellipse` draws ellipses around groups



Scatterplot Annotations

```
1 penguins |> ggplot(aes(x=body_mass_g, y=flipper_length_mm, color=species, shape=species))
2   geom_point() +
3   stat_ellipse(size=0.5) +
4   other_commands(...)
```



Labeling Ellipses

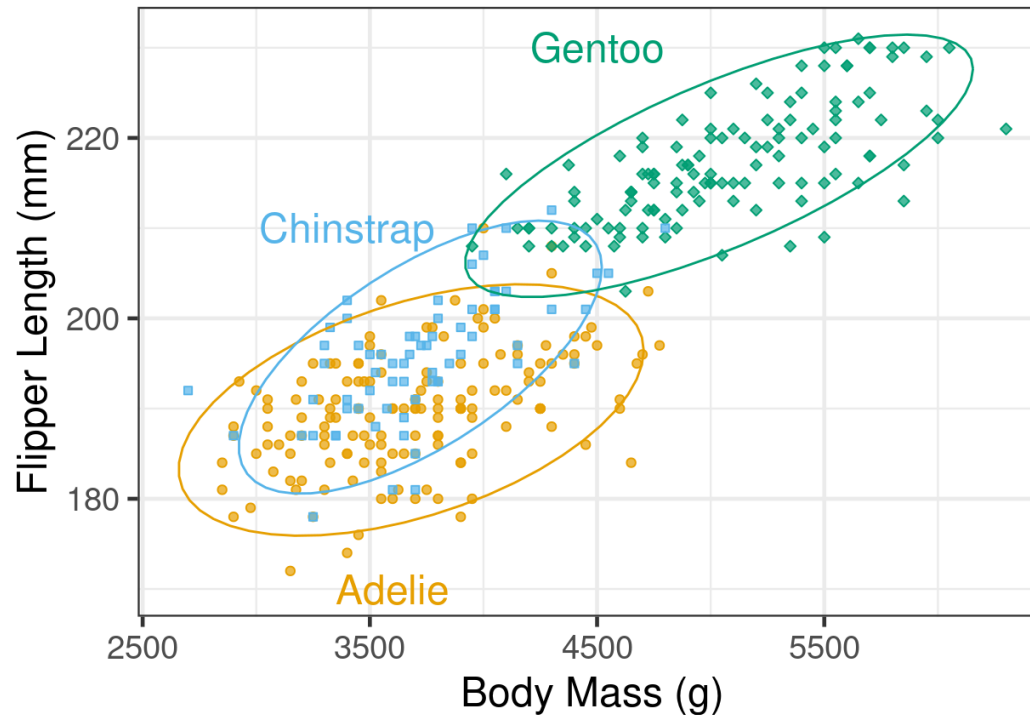
- Can use `geom_text` to label ellipses
- Create a new data-frame for the labels

```
1 penguin_labels = tibble(  
2   species = c("Adelie", "Chinstrap", "Gentoo"),  
3   body_mass_g = c(3600, 3400, 4500),  
4   flipper_length_mm = c(170, 210, 230),  
5   hjust = c(0, 0.5, 0),  
6   vjust = c(0, 0.5, 1)  
7 )
```

Labelling Ellipses

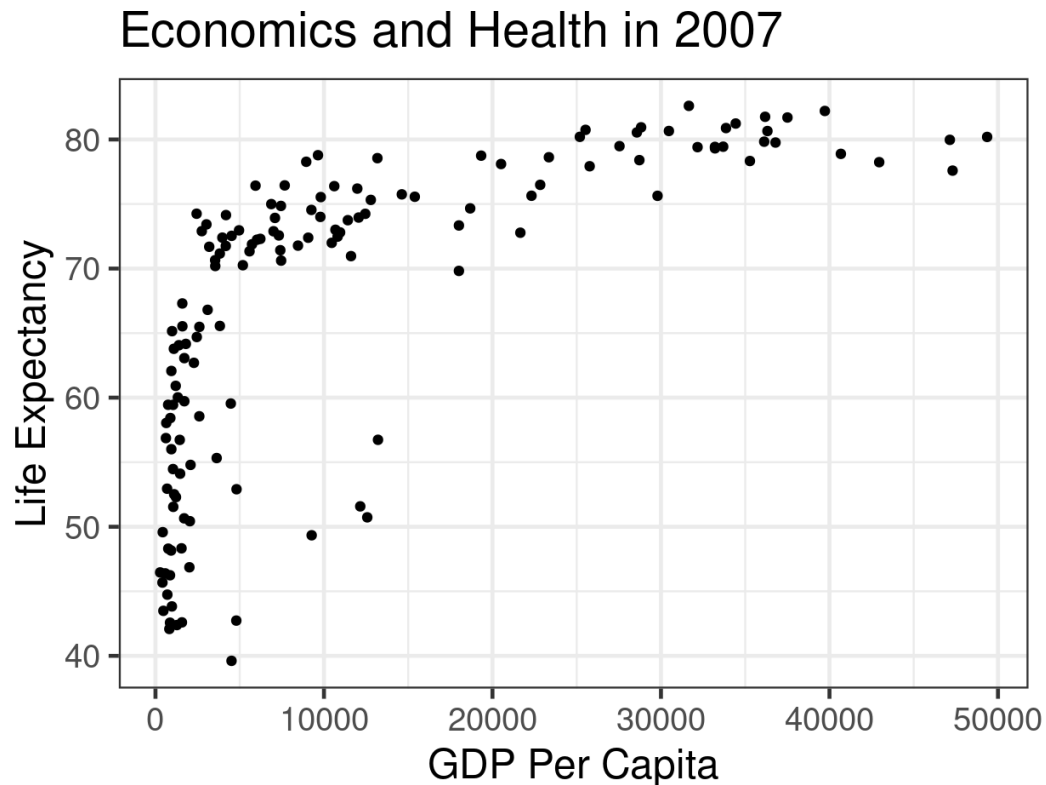
```
1 p1 +  
2   geom_text(data = penguin_labels,  
3             label = penguin_labels$species,  
4             size = 18/.pt) +  
5   theme(legend.position = "none")
```

Ellipses Clarify



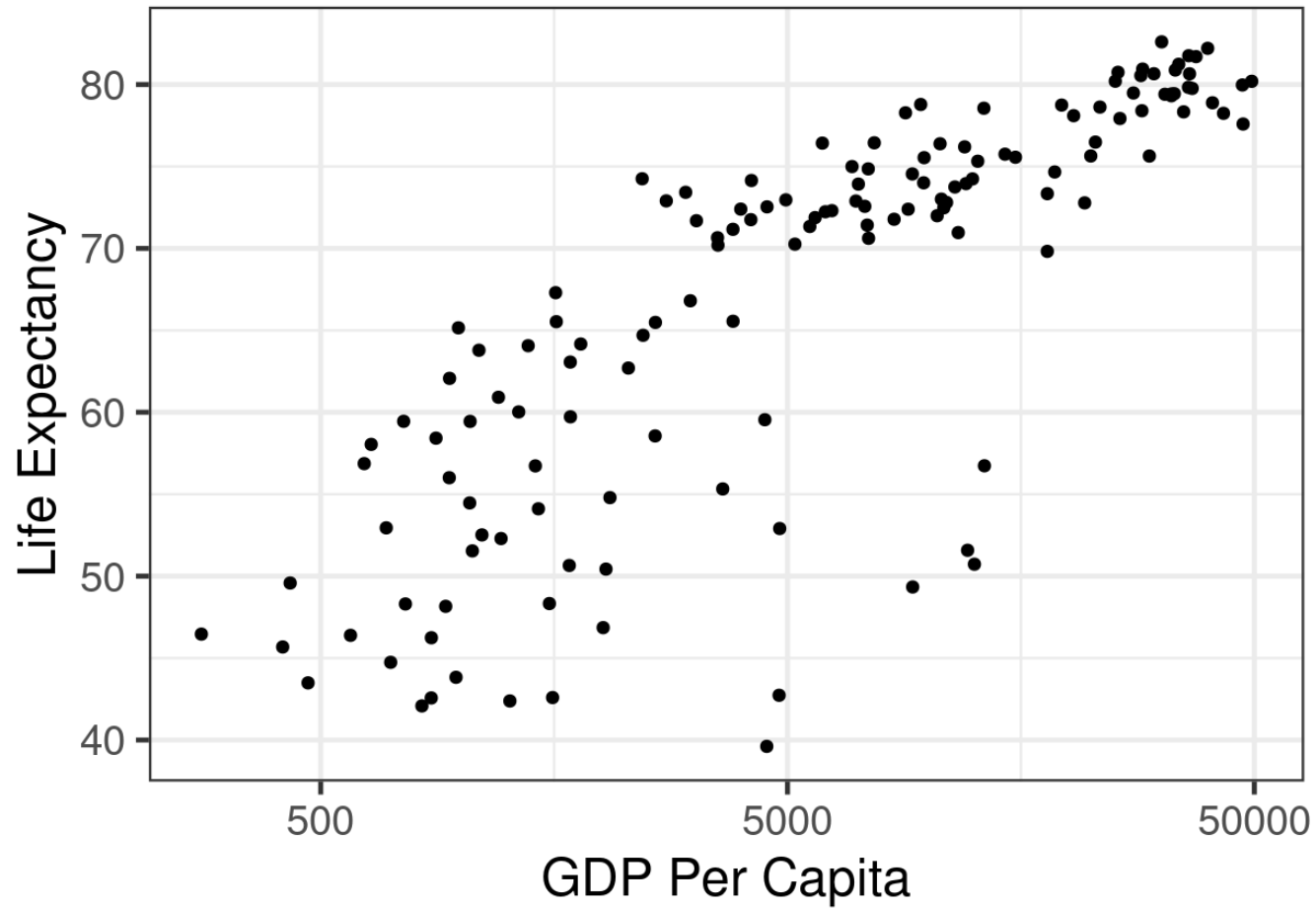
Unique Elements

- When the category elements are unique makes sense to label them with points



log scale

Economics and Health in 2007



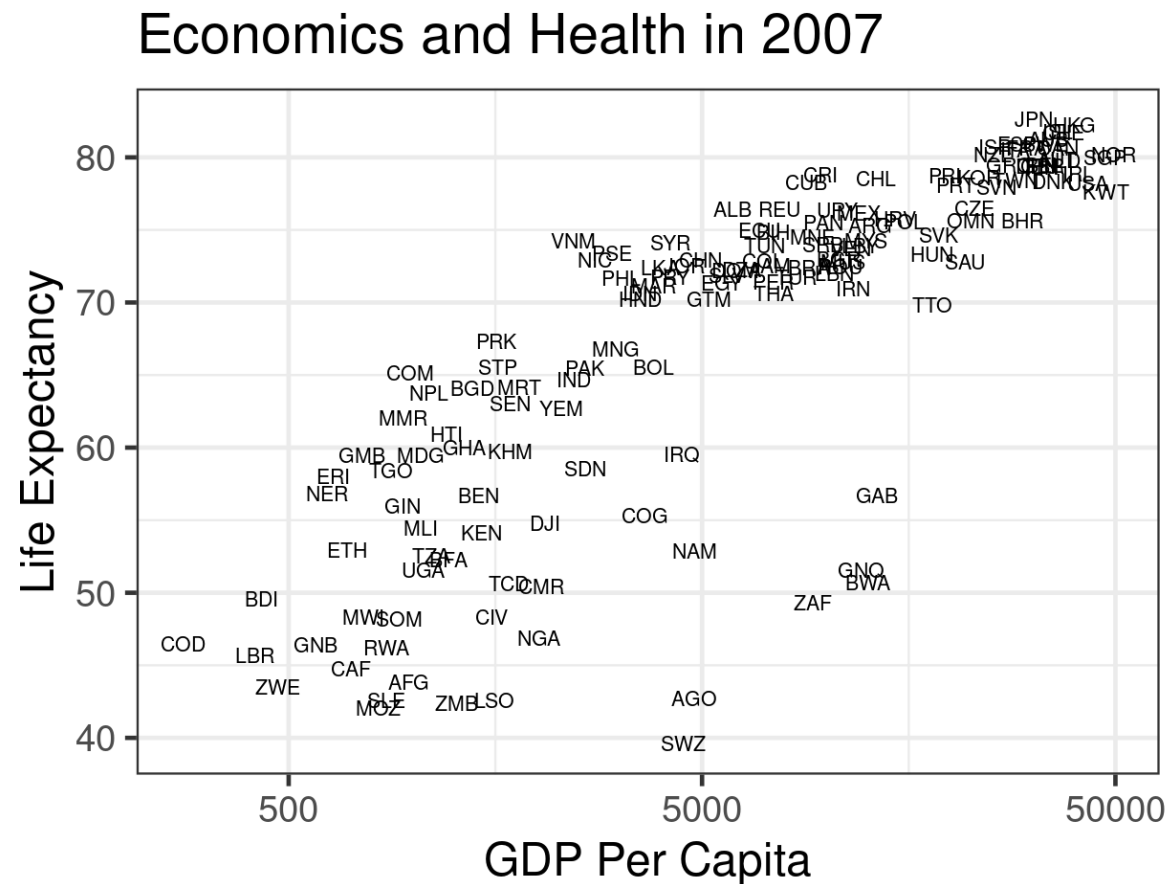
log scale

```
1 gapminder |> filter(year == 2007) |> ggplot(aes(x=gdpPercap,y=lifeExp))+  
2   geom_point() +  
3   theme_bw(base_size = 18) +  
4   theme(aspect.ratio=0.67) +  
5   xlab("GDP Per Capita") +  
6   ylab("Life Expectancy") +  
7   labs(title = "Economics and Health in 2007") +  
8   scale_x_log10(breaks = c(500, 5000, 50000))
```

- **log** is useful when data is positive and spans a large range
- **scale_x_log10** and **scale_y_log10** can be used

Adding labels

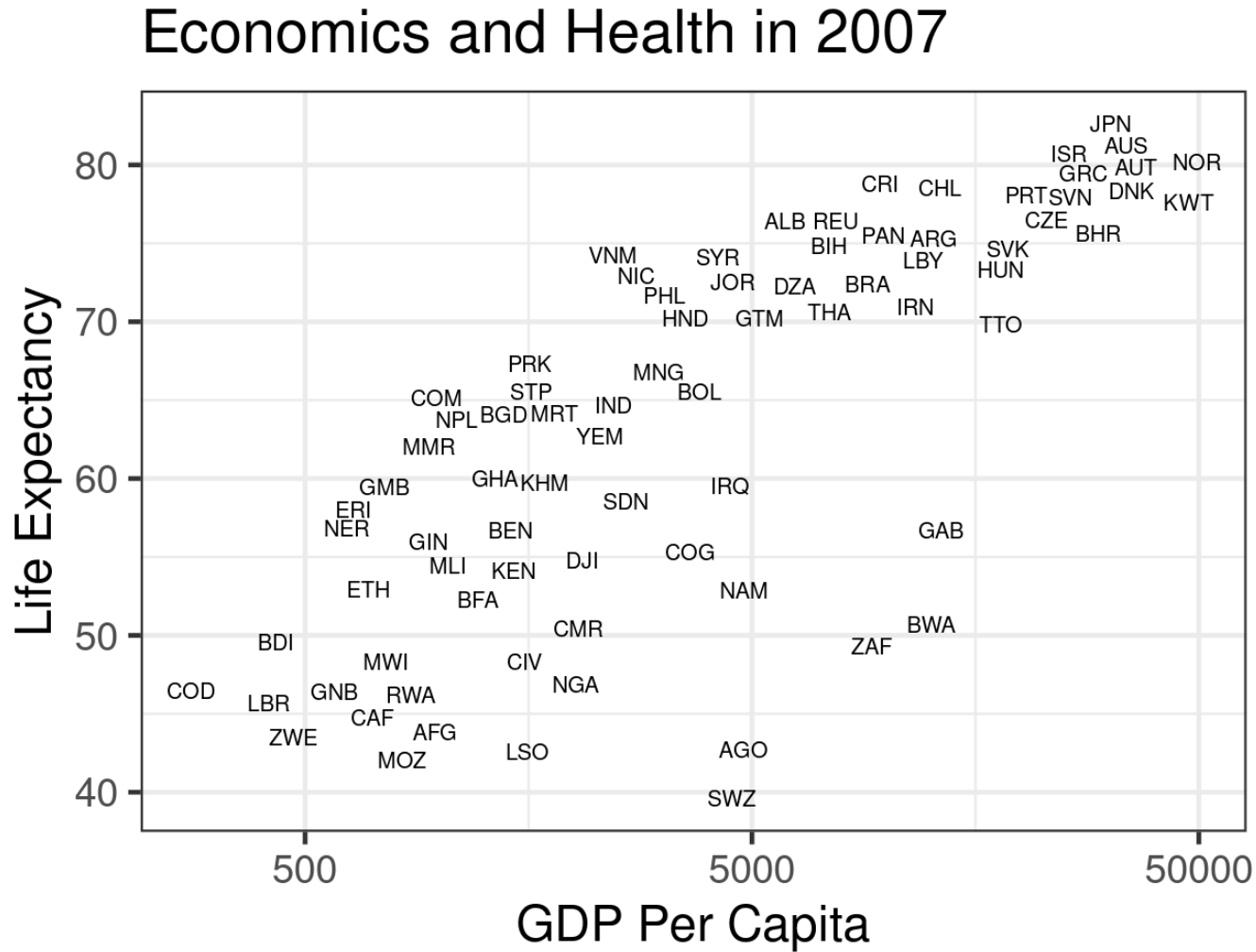
- Can use `geom_text` instead of `geom_point`



Adding Labels

```
1 gapminder |> left_join(country_codes) |>
2   filter(year == 2007) |> ggplot(aes(x=gdpPercap,y=lifeExp))+
3   geom_text(aes(label = iso_alpha),size=8/.pt) +
4   theme_bw(base_size = 18) +
5   theme(aspect.ratio=0.67) +
6   xlab("GDP Per Capita") +
7   ylab("Life Expectancy") +
8   labs(title = "Economics and Health in 2007") +
9   scale_x_log10(breaks = c(500, 5000, 50000))
```

geom_text removing overlaps



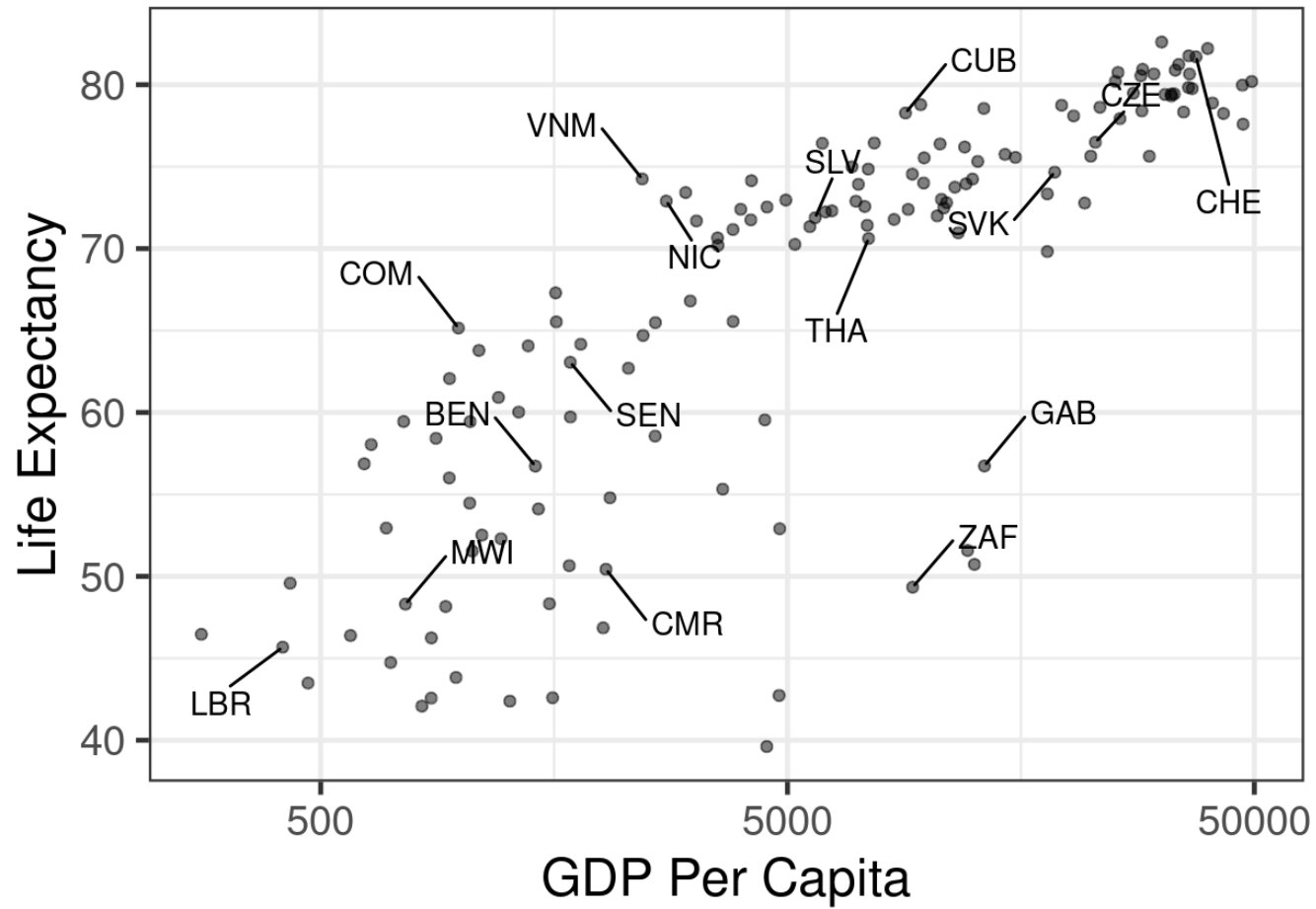
geom_text removing overlaps

```
1 gapminder |>
2   left_join(country_codes) |>
3   filter(year == 2007) |>
4   ggplot(aes(x=gdpPercap,y=lifeExp))+
5   geom_text(aes(label = iso_alpha),
6             size=8/.pt,
7             check_overlap = TRUE) +
8   theme_bw(base_size = 18) +
9   theme(aspect.ratio=0.67) +
10  xlab("GDP Per Capita") +
11  ylab("Life Expectancy") +
12  labs(title = "Economics and Health in 2007") +
13  scale_x_log10(breaks = c(500, 5000, 50000))
```

- But throws away data randomly

Label a subset

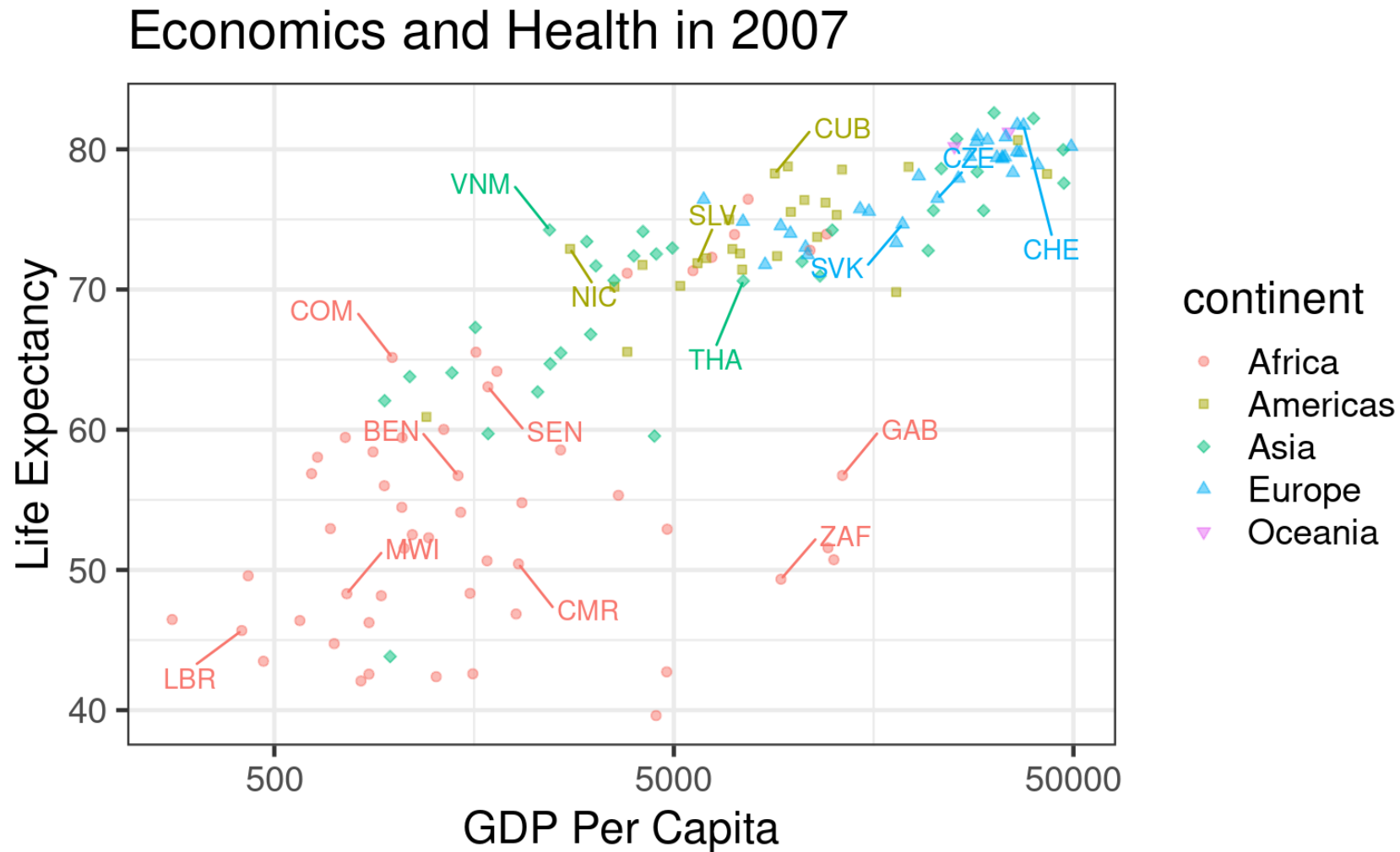
Economics and Health in 2007



Label a subset

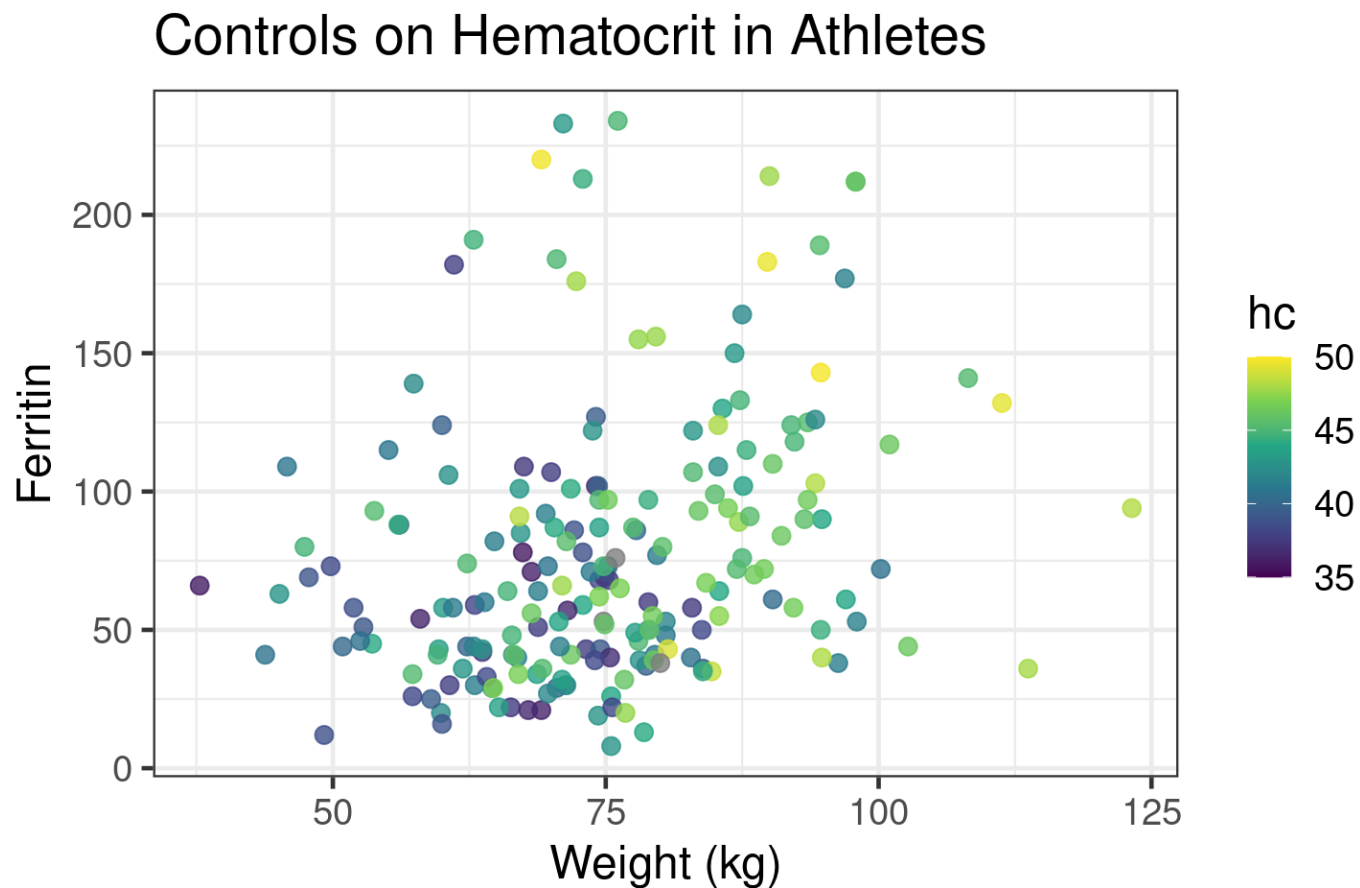
```
1 library(ggrepel)
2
3 set.seed(1000)
4 label_list = country_codes |> slice_sample(n=20) |> pull(iso_alpha)
5 gapminder_2 = gapminder |> left_join(country_codes) |>
6   mutate(iso_alpha = if_else(iso_alpha %in% label_list, iso_alpha, NA))
7
8 gapminder2 |>
9   filter(year == 2007) |>
10  ggplot(aes(x=gdpPercap, y=lifeExp)) +
11    geom_point(alpha=0.5) +
12    geom_text_repel(aes(x=gdpPercap,
13                        y=lifeExp,
14                        label = iso_alpha),
15                  max.overlaps=Inf,
16                  min.segment.length = 0.0,
17                  box.padding = 1) + ...
```

Add a category



Color/Bubble Plots

- With 3 quantitative variables you can use size or color



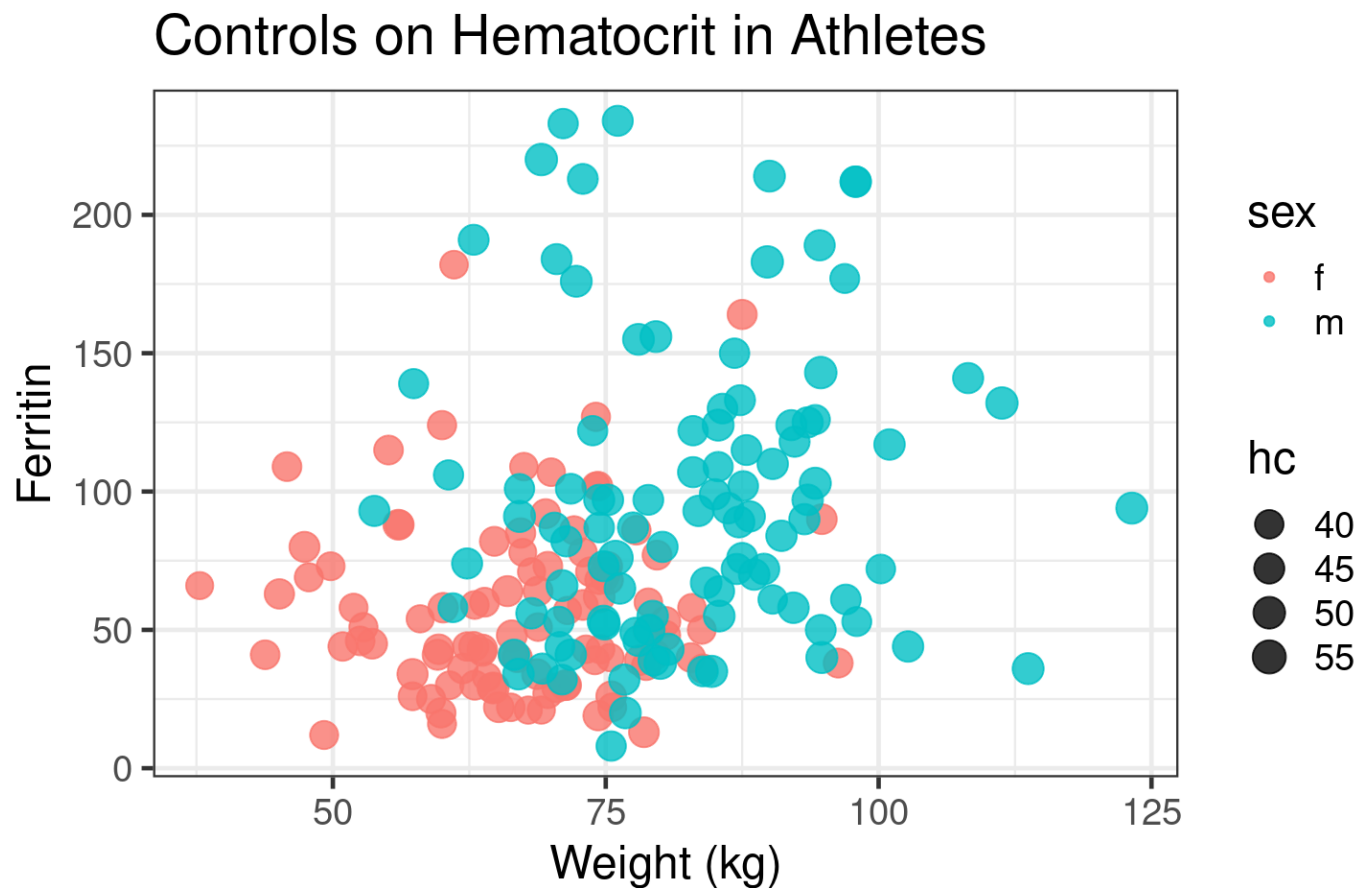
Color/Bubble Plots

- With 3 quantitative variables you can use size or color

```
1 library(ggribes)
2 Aus_athletes |> ggplot(aes(x=weight,y=ferr,color=hc)) +
3   geom_point(size=3,alpha=0.8) +
4   scale_color_viridis_c(limits=c(35,50)) +
5   theme_bw(base_size = 18) +
6   theme(aspect.ratio = 0.67) +
7   ylab("Ferritin") +
8   xlab("Weight (kg)") +
9   labs(title = "Controls on Hematocrit in Athletes")
```


Bubble Plot

- The Bubble plot uses size instead

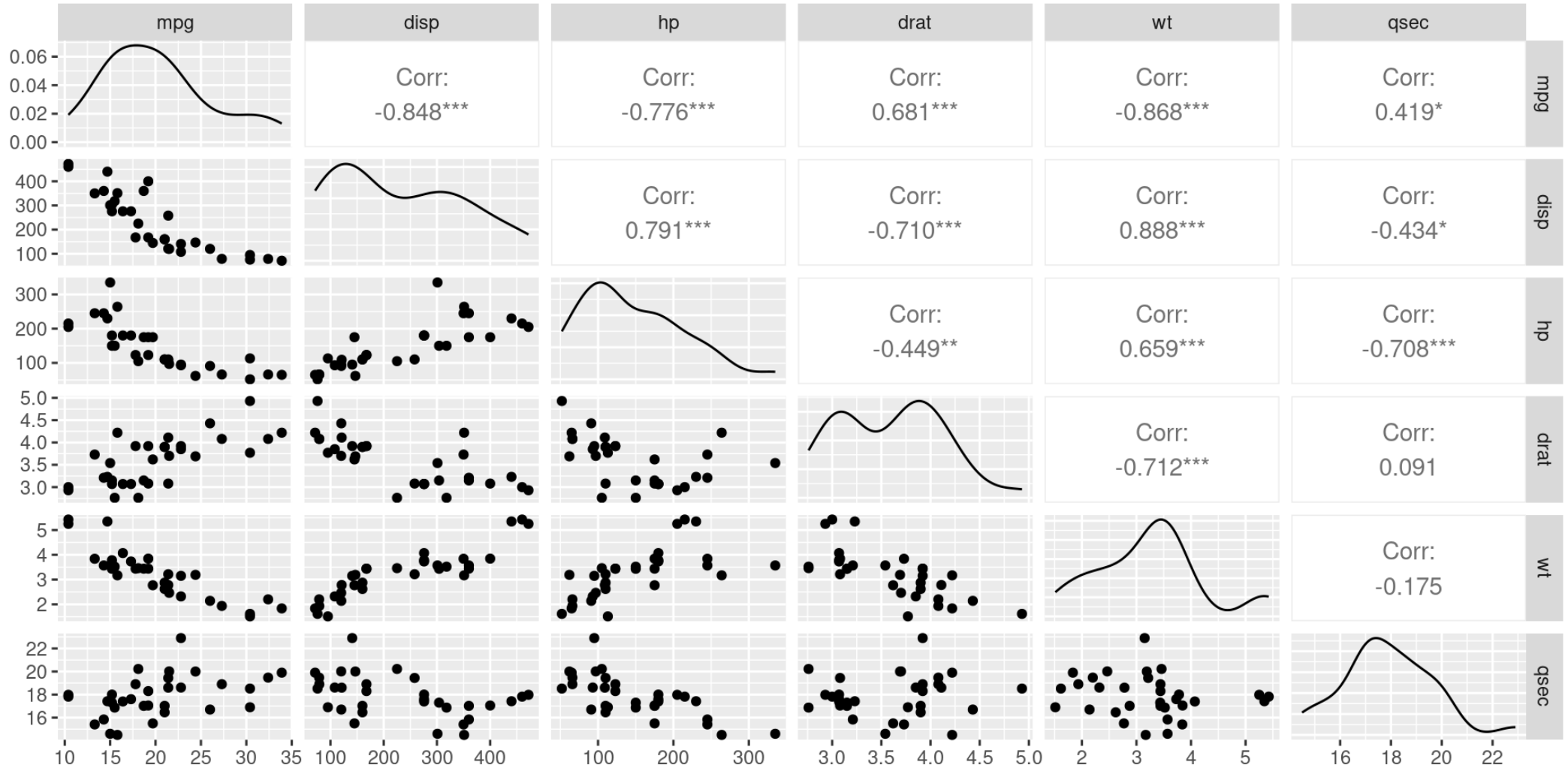


Bubble Plot

- The Bubble plot uses size instead

```
1 library(ggribes)
2 Aus_athletes |> ggplot(aes(x=weight,y=ferr,size=hc,color=sex,fill=sex)) +
3   geom_point(alpha=0.8) +
4   scale_size_area() +
5   theme_bw(base_size = 18) +
6   theme(aspect.ratio = 0.67) +
7   ylab("Ferritin") +
8   xlab("Weight (kg)") +
9   labs(title = "Controls on Hematocrit in Athletes")
```

Many Variables: Pair-Plots



Many Variables: Pair-Plots

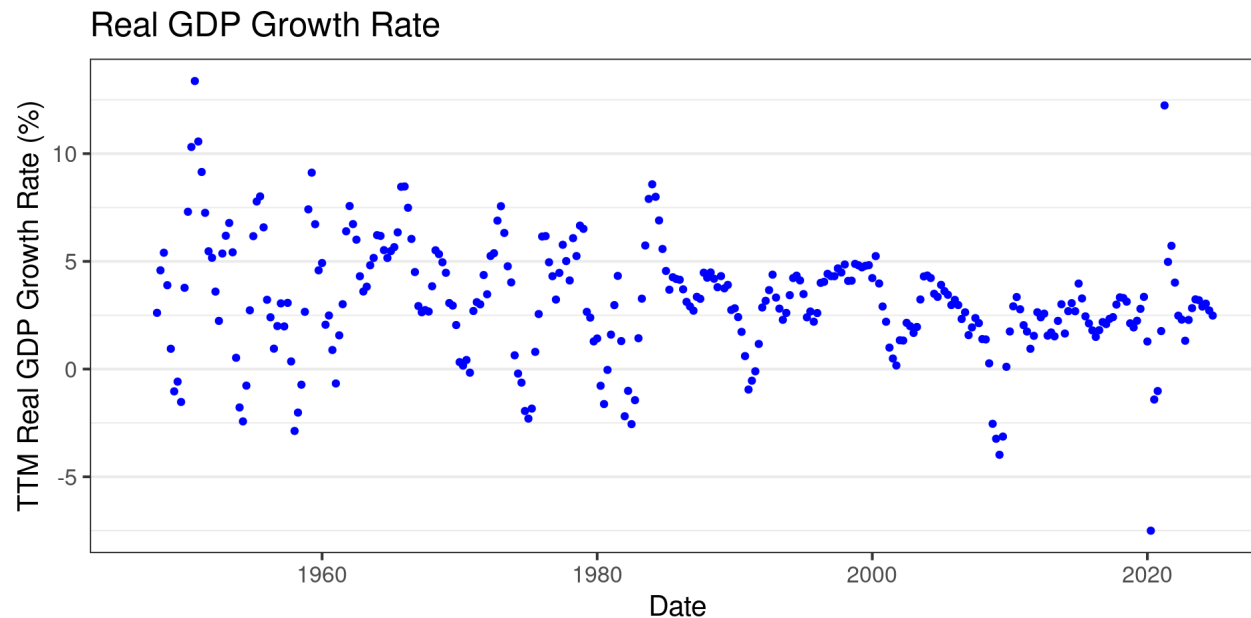
```
1 library(GGally)
2
3 mtcars |> select(mpg, disp, hp, drat, wt, qsec) |> ggpairs()
```

Disadvantages of Pair-Plots

- When the number of variables is high, false discovery rate
- Can be challenging to look at
- Later in semester we will discuss dimensional reduction

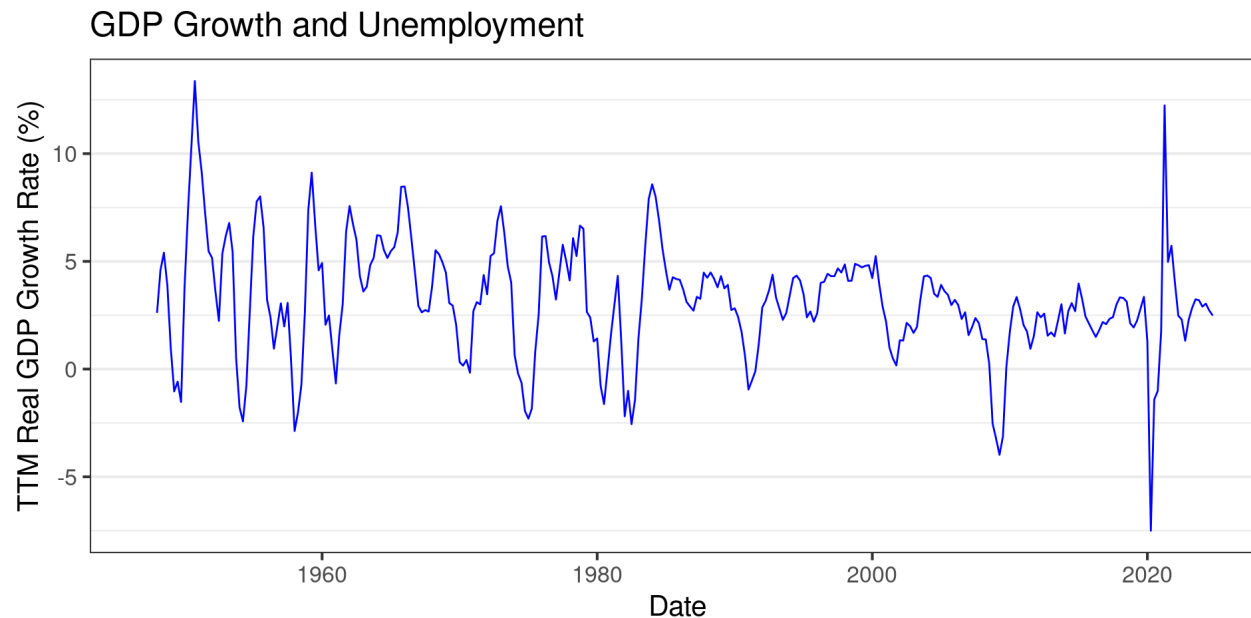
Time Series

- Time series are like a scatterplot
- Time variable gives additional structure
- Commonly use line plots instead of points



Time Series

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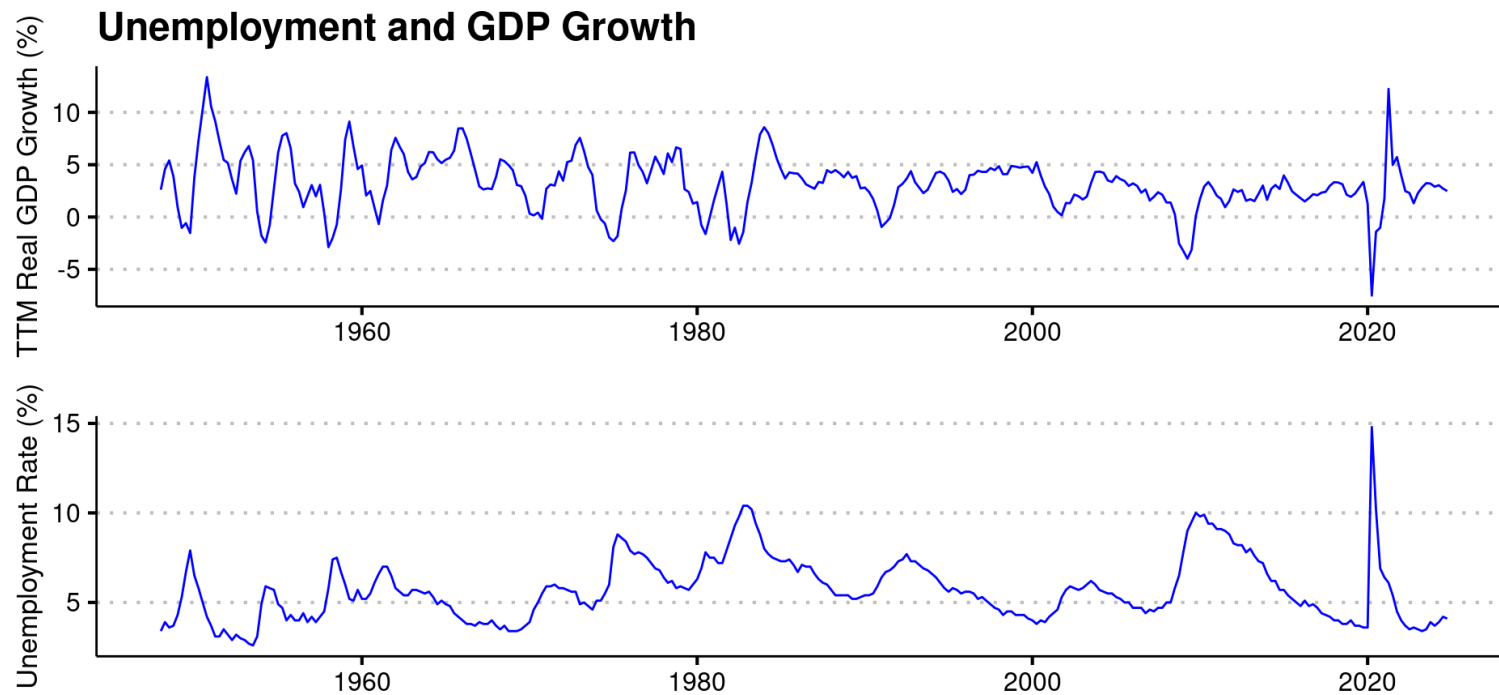
Time Series

- Time series are like a scatterplot
- Time variable gives additional structure
- Commonly use line plots instead of points

```
1 library(TTR)
2 gdp = read_csv("GDPC1.csv")
3 gdp = gdp |> mutate(growth = 100*(GDPC1/lag(GDPC1,4) - 1))
4
5 gdp |> ggplot(aes(x=observation_date,y=growth)) +
6   geom_line(color="blue") +
7   theme_bw(base_size=16) +
8   theme(panel.grid.minor.x = element_blank(),
9         panel.grid.major.x = element_blank() )+
10  labs(title="Real GDP Growth Rate") +
11  xlab("Date") +
12  ylab("TTM Real GDP Growth Rate (%)")
```


Multiple Time Series

- Unless units are the same, use multiple plots
- I like to use the `patchwork` package



Multiple Time Series

- Unless units are the same, use multiple plots
- I like to use the `patchwork` package

```
1 library(patchwork)
2 unemp = read_csv("UNRATE.csv")
3 p1 = gdp |> ggplot(aes(x=observation_date,y=growth)) +
4   geom_line(color="blue") +
5   theme_clean(base_size=16) +
6   theme(panel.grid.minor.x = element_blank(),
7         panel.grid.major.x = element_blank(),
8         plot.background = element_rect(color = NA)
9         ) +
10  labs(title = "Unemployment and GDP Growth")+
11  xlab("") +
12  ylab("TTM Real GDP Growth (%)")
13
14 p2 = unemp |> ggplot(aes(x=observation_date,y=UNRATE)) +
15   geom_line(color="blue") +
16   theme_clean(base_size=16) +
17   theme(panel.grid.minor.x = element_blank(),
```

18

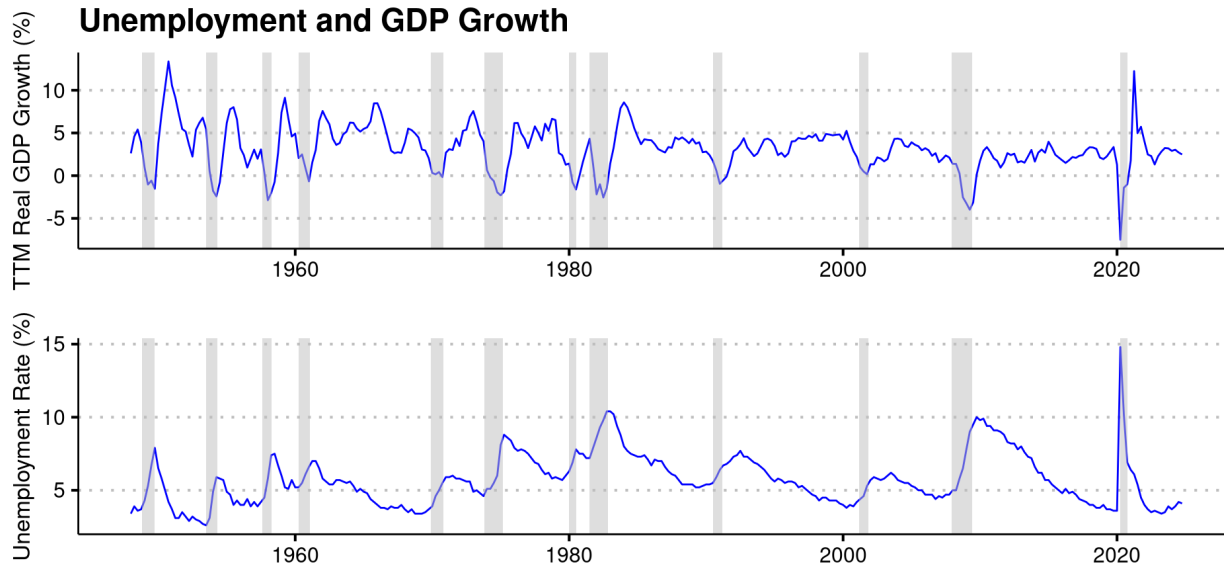
```
panel.grid.major.x = element_blank(),
```

19

```
panel.grid.minor.x = element_blank(),
```

How to Annotate

- Identify recessions with shaded rectangles
- `annotate` function takes `geom` as argument
- Here we identified recessions using FRED data



How to Annotate

- Identify recessions with shaded rectangles
- `annotate` function takes `geom` as argument
- Here we identified recessions using FRED data

```
1 recessions = read_csv("recessions.csv")
2 recessions = recessions |> filter(Peak>"1948-01-01")
3
4 p1_shade = p1 +
5   annotate("rect",xmin=recessions$Peak,xmax=recessions$Trough,ymin=-Inf,yma
6
7 p2_shade = p2 + annotate("rect",xmin=recessions$Peak,xmax=recessions$Trough
8 p1_shade/p2_shade
```

Avoiding the Legend

- Create second data-frame with the last values
- Add additional “axis” on right hand side

```
1 tech_stocks = read_csv("../Meetup3/tech_stocks.csv")
2
3 p = tech_stocks |>
4   ggplot(aes(x=date,y=return,group=Company,color=Company)) + geom_line(size
5   geom_hline(yintercept = 1, size = 1.0, color="grey70") +
6   theme_gray(base_size = 24) +
7   scale_color_manual(
8     values = c( "#009E73", "#E69F00", "#000000", "#56B4E9"),
9     name = NULL,
10    breaks = c("NVDA", "META", "AAPL", "MSFT"),
11    labels = c("NVidia", "Meta", "Apple", "Microsoft")) +
12  theme(
13    strip.background = element_blank(),
14    panel.background = element_rect(fill = "#E6E6E3",
15    linewidth = 0),
16    panel.grid.minor.y = element_line(color="white",linewidth = 0.5),
17    panel.grid.major.y = element_line(color="white",linewidth = 0.5),
```

18

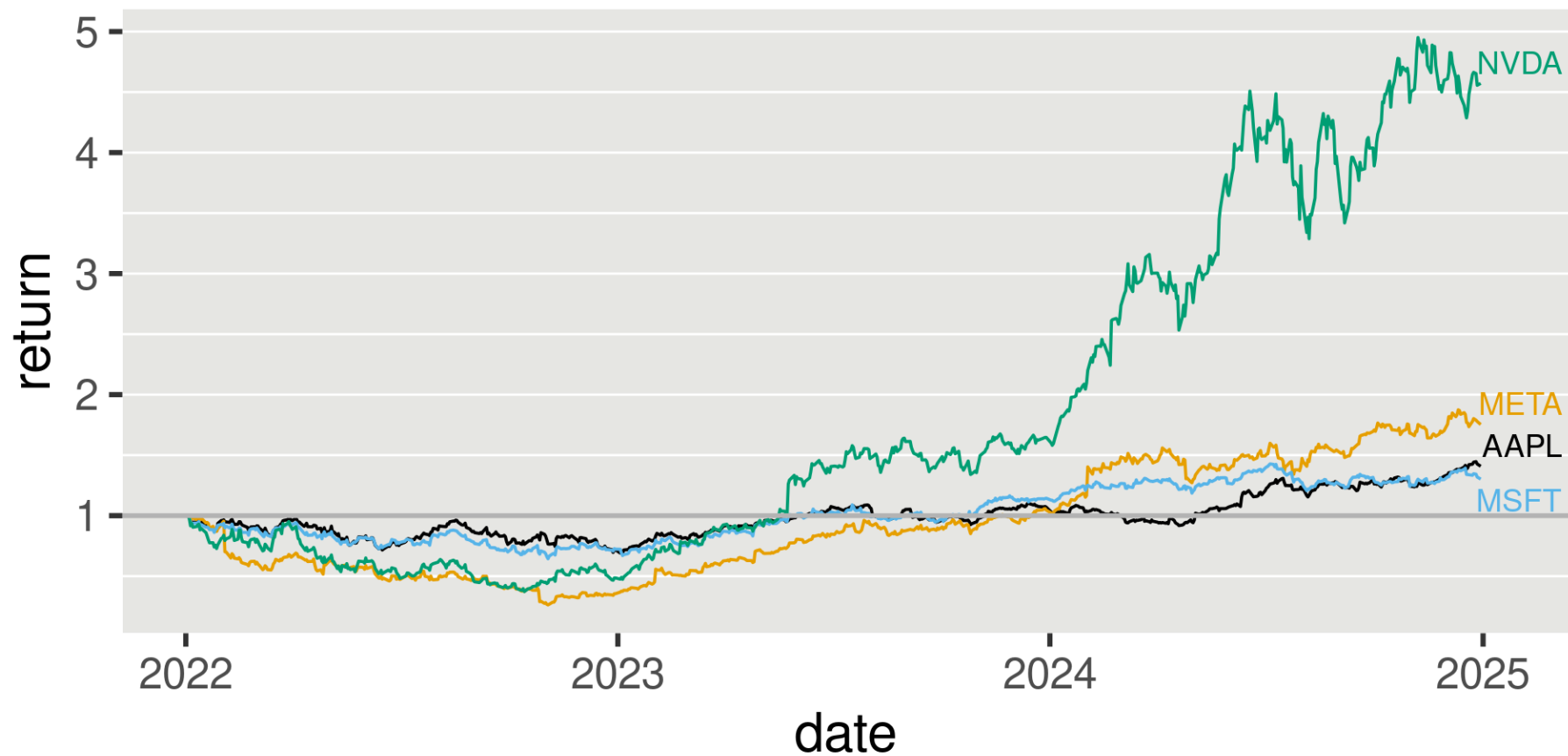
```
panel.grid.minor.x = element_blank(),
```

19

```
  facet_grid(. ~ ., scales = 'free',
```

Avoiding the Legend

- Create second data-frame with the last values



Avoiding the Legend

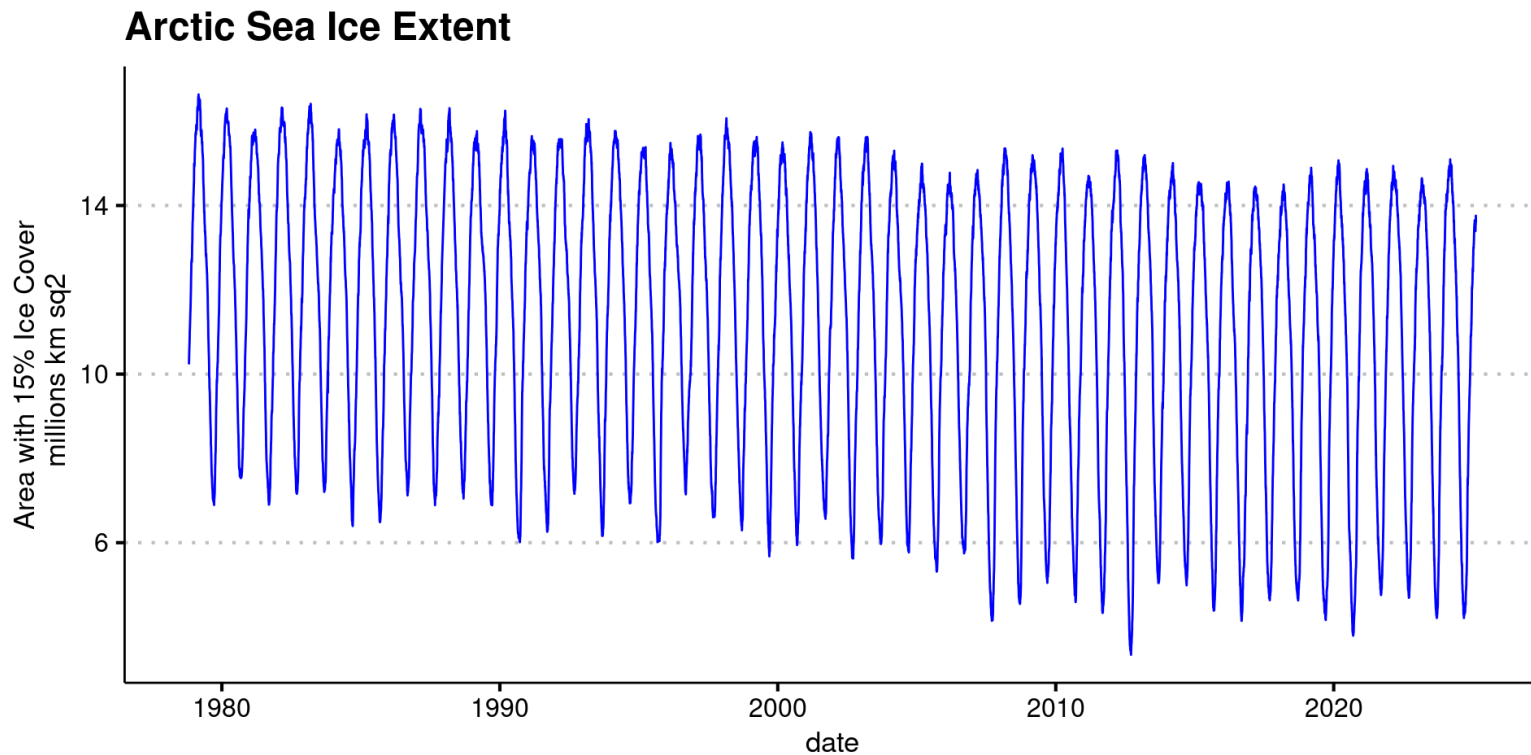
- Create second data-frame with the last values

```
1 tech_stocks_final = tech_stocks |> filter(date == max(date))
2 p + geom_text_repel(data=tech_stocks_final, aes(x=date, y=return, label=Compan
3   guides(color = "none", size="none"))
```

- Another technique involves creating a secondary axis with breaks at the final returns

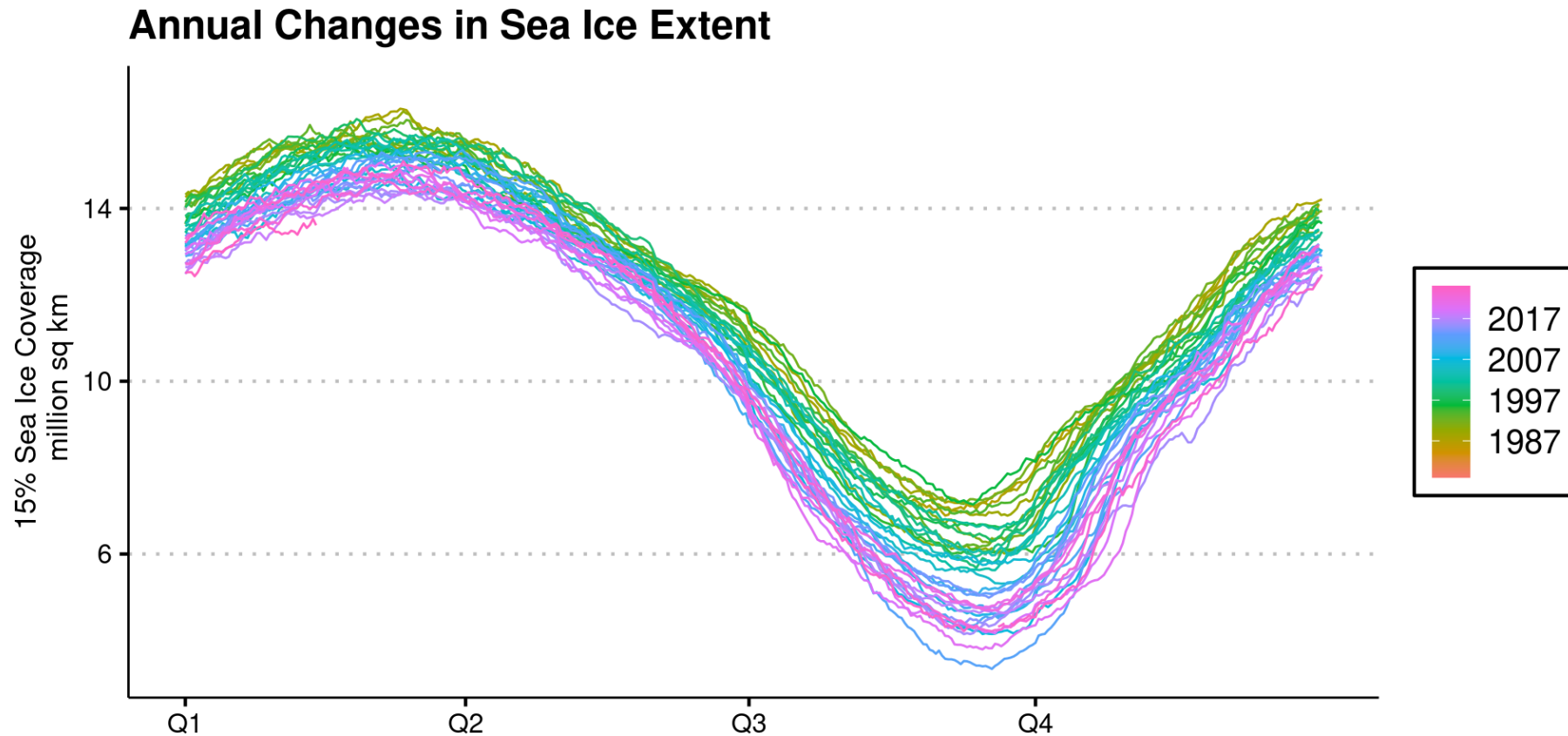
Seasonal and Polar Plots

- Lots of time series data show annual or daily patterns
- Plot against just month/day or in polar coordinates



Seasonal and Polar Plots

- use `feasts` package



Seasonal and Polar Plots

- use `feasts` package

```
1 library(feasts)
2 sea_ice |>
3   as_tsibble() |>
4   tsibble::fill_gaps() |>
5   gg_season(Extent) +
6   theme_clean(base_size=16) +
7   theme(plot.background = element_rect(color = NA)) +
8   labs(title = "Annual Changes in Sea Ice Extent") +
9   xlab("") +
10  ylab("15% Sea Ice Coverage\n million sq km")
```

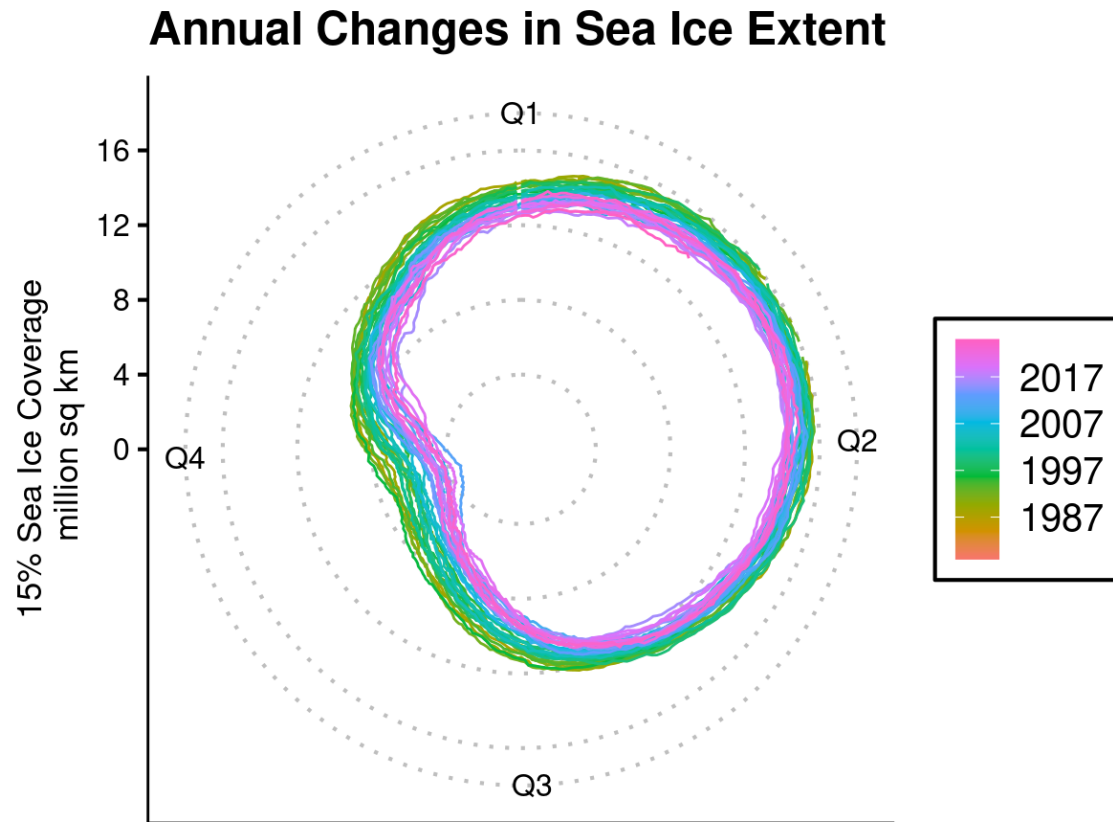
Seasonal and Polar Plots

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Seasonal and Polar Plots

- use `feasts` package



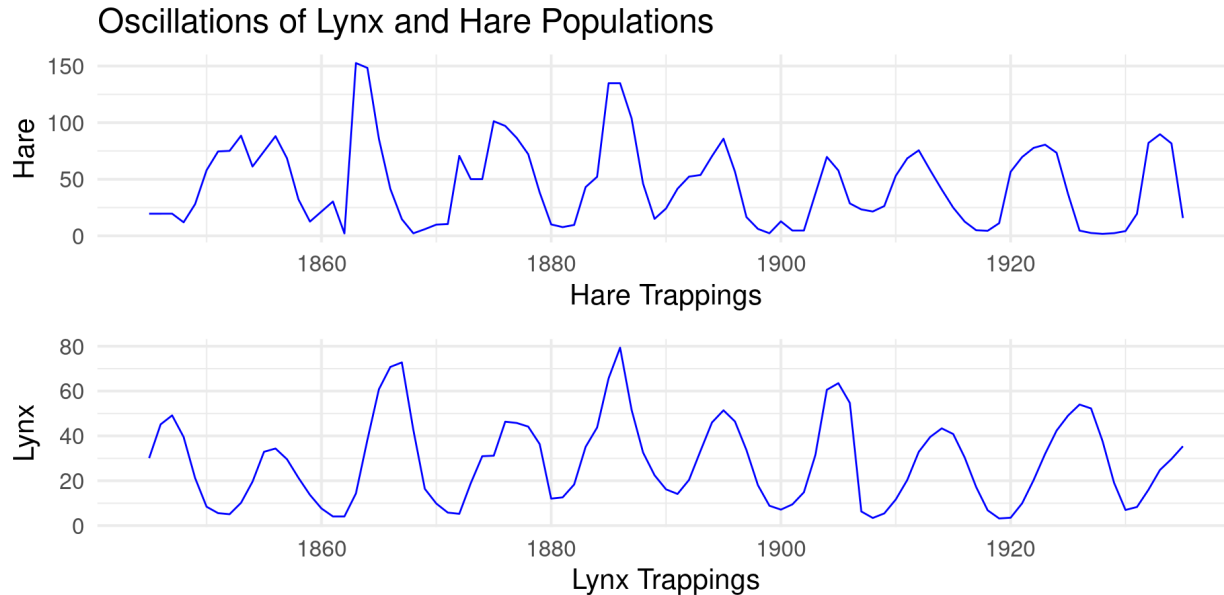
Seasonal and Polar Plots

- use `feasts` package

```
1 library(feasts)
2 sea_ice |>
3   as_tsibble() |>
4   tsibble::fill_gaps() |>
5   gg_season(Extent, polar=TRUE) +
6   theme_clean(base_size=16) +
7     theme(plot.background = element_rect(color = NA)) +
8   labs(title = "Annual Changes in Sea Ice Extent") +
9   scale_y_continuous(limits = c(0,16)) +
10  xlab("")+
11  ylab("15% Sea Ice Coverage\n million sq km")
```

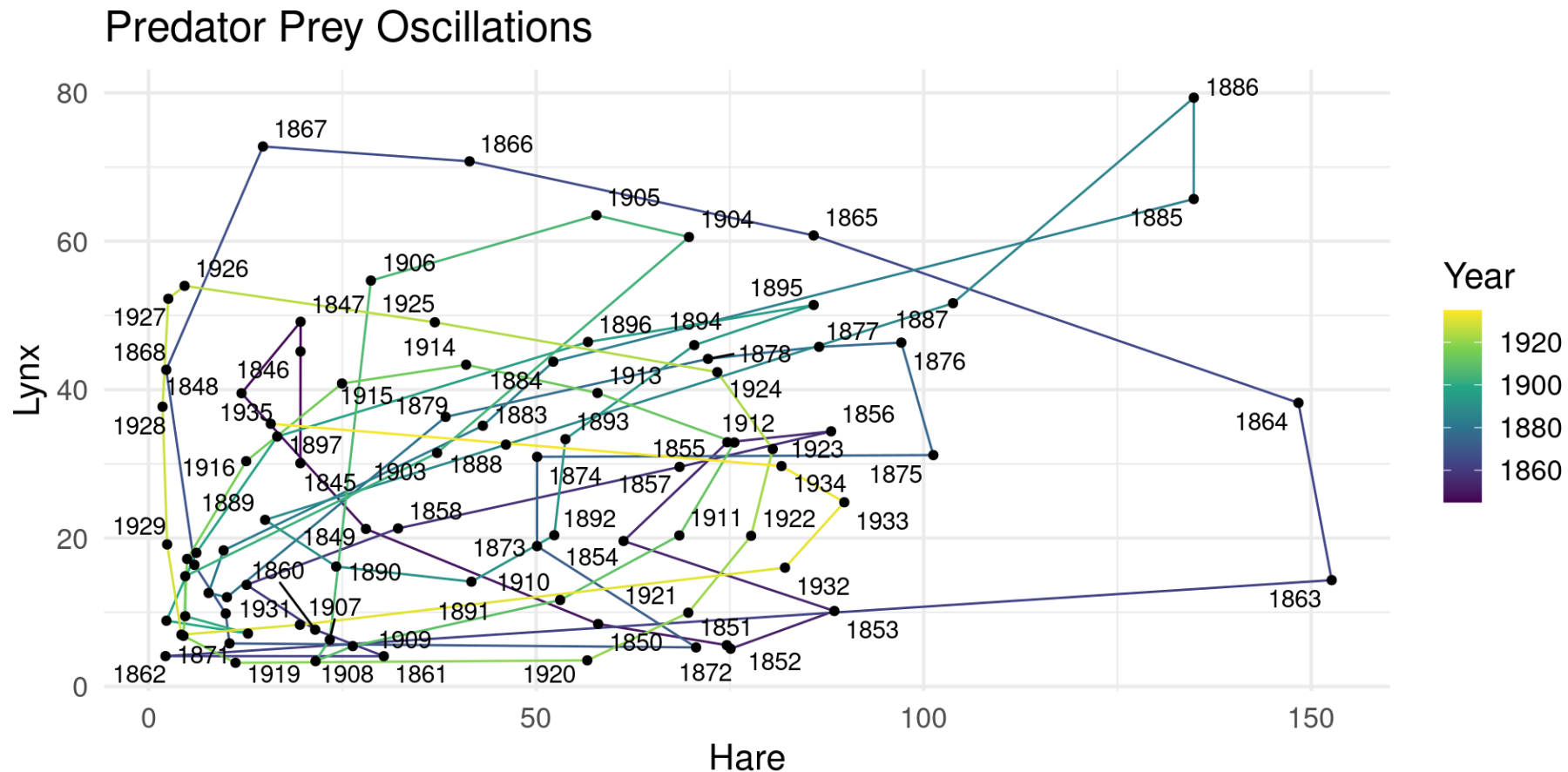
Connected Scatterplot

- Alternative way to show two time series together
- Inspired by “phase-portrait” in physics
- `geom_path`



Connected Scatterplot

- Alternative way to show two time series together



Connected Scatterplot

- Alternative way to show two time series together

```
1 hare_lynx |> ggplot(aes(x=Hare,y=Lynx)) +  
2   geom_path(aes(color=Year)) +  
3   geom_point() +  
4   geom_text_repel(aes(label = Year, x = Hare, y = Lynx)) +  
5   scale_color_viridis_c() +  
6   theme_minimal(base_size = 16) +  
7   theme(plot.background = element_rect(color = NA)) +  
8   labs(title="Predator Prey Oscillations") +  
9   xlab("Hare") +  
10  ylab("Lynx")
```

Connected Scatterplot

The New York Times

METRICS

HANNAH FAIRFIELD

Driving Shifts Into Reverse

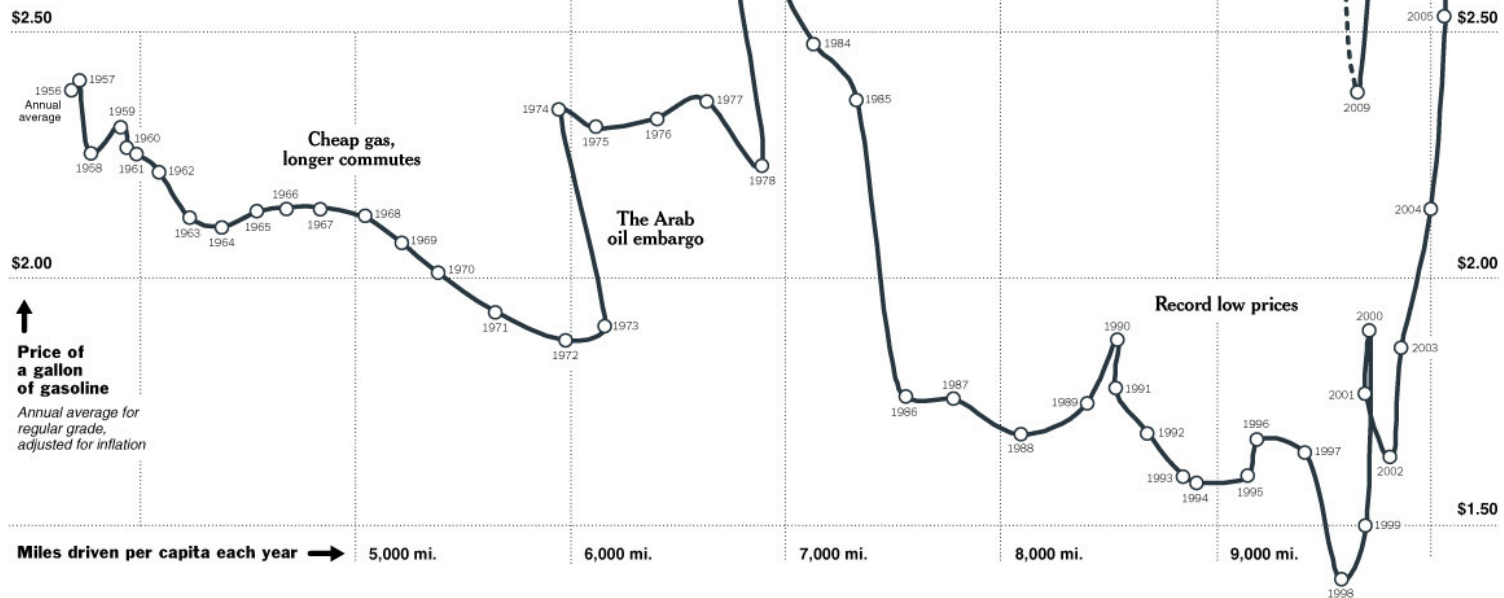
ECONOMISTS have long studied the relationship between driving habits and gasoline prices. Low gas prices can bring periods of profligate driving, and a quick jump in prices can cause many vehicles to languish in garages.

Until recently, Americans have driven more each year than the previous one, with a few brief exceptions. In 1956, Americans of driving age drove about 4,000 miles a year, on average. Fifty years later, that figure had climbed above 10,000.

But the latest recession has caused some big changes. High unemployment meant that fewer people were driving to work, and a slump in consumer spending

meant that less freight needed to be moved around the country. As gas prices soared in 2005, the number of miles driven — including commercial and personal — began to fall, and continued to drop after 2008 even as gasoline became cheaper.

"People were surprised by the very rapid rise in gas prices, and they changed their driving behavior," said Kenneth A. Small, a transportation economist at the University of California, Irvine. "But my suspicion is that it is temporary. As soon as unemployment gets back to pre-recession levels, we will see Americans doing a lot more driving again."



NY Times 2010

Thanks